

MINISTRY OF HOUSING AND URBAN AFFAIRS**NOTIFICATION**

New Delhi, the 16th December, 2020

G.S.R. 769(E).—In exercise of the powers conferred by clause (c) of sub-section (2) of section 22 and clause (e) of sub-section (2) of section 100 of the Metro Railways (Operation and Maintenance) Act, 2002 (60 of 2002) and in supersession of the Metro Railways General Rules, 2013, except as respects things done or omitted to be done, the Central Government hereby makes the following rules for regulating the operations and maintenance of metro railways in India, except the metropolitan city of Calcutta, namely:-

CHAPTER I**PRELIMINARY****1. Short title and commencement—**

- (1) These rules may be called the Metro Railways General Rules, 2020.
- (2) These rules shall come into force on the date of their publication in the Official Gazette.

2. Definitions —

- (1) In these rules, unless the context otherwise requires, —
 - (i) “accident” means any occurrence which causes or has the potential to cause death or injury to staff, passengers or other persons or cause damage to the property of the metro railway, passengers or other persons;
 - (ii) “Act” means the Metro Railways (Operation and Maintenance) Act, 2002, (60 of 2002);
 - (iii) “adequate distance” means the distance sufficient to ensure safety;
 - (iv) “approach lighting” means an arrangement in which the lighting of signals is controlled automatically by the approach of a train;
 - (v) “approved special instructions” means special instructions approved by the Commissioner;
 - (vi) “authorised officer” means an officer who is duly empowered by general or special order of the metro railway administration, either by name or by virtue of his office to issue instructions;
 - (vii) “authorised electricalperson” means aperson who is duly authorised by an officer of the metro railway administration,either by name or by designation to perform specific work on the electrical equipment or circuitry;
 - (viii) “authorisedperson” means anemployee or a person to whom a competency certificate has been issued by the metro railway administration;
 - (ix) “authority to proceed” means the authority given to the operator of a train or UTO train, under the system of workingto enter the block section with his train;
 - (x) “automatic fare collection system” means an automatic system for collection of fares and issuing of tickets;
 - (xi) “Automatic Mode (AM)” means the mode of operation of train under Automatic Train Operation where train is driven automatically including control of acceleration, coasting, braking and stopping of trains;
 - (xii) “Automatic Train Operation (ATO)” means a sub-system of continuous automatic train control system, which automatically controls acceleration, coasting, braking and stopping of trains;
 - (xiii) “Automatic Train Protection (ATP)” means a sub-system of continuous automatic train control system which maintains safe train operation, including train direction, train separation, interlocking and speed enforcement;

- (xiv) “Automatic Train Supervision (ATS)” means a sub-system of continuous automatic train control system which automatically monitors the entire system and directs train running so as to provide scheduled service under normal circumstances;
- (xv) “auxiliary systems controller” means an authorised person responsible for the control of the auxiliary systems including tunnel ventilation system, station air conditioning and building management system of the metro railway;
- (xvi) “axle counter” means an electrical device which, when provided at two given points on the track, proves, by counting ‘axles in’ and counting ‘axles out’, whether the section of the track between the said two points is clear or occupied;
- (xvii) “baggage handling system” means a mechanised train borne and station-based system meant for transfer of passenger’s baggage between stations and to the airport terminal;
- (xviii) “berth” means length of track nominated to be occupied by a train adjacent to a platform or in the depot;
- (xix) “block section” means the portion of the running line, as specified by special instructions, on to which no running train may enter, until permission to approach has been received;
- (xx) “blue light station” means a location indicated by a blue light where a person can communicate with the OCC and arrange to disconnect traction power of a specified section, in an emergency;
- (xxi) “building management system” means a control-panel or workstation installed at every station to monitor and control proper functioning of building services, including safety critical systems;
- (xxii) “cab signal” means visual indication displayed as speed limit and target distance on the Train Operator’s console granting him the authority to proceed under automatic mode, or coded manual mode of driving;
- (xxiii) “calendar day” means the period from midnight of one day to midnight of subsequent day;
- (xxiv) “car shed” or “service depot” means an area where the metro railway trains and coaches are berthed either for repair or for any other attention including stabling;
- (xxv) “caution order” means an instruction given to the Train Operator to observe special precautions including speed reduction at notified locations;
- (xxvi) “certificate of competency” means the certificate issued to a person after he has been examined for his knowledge of rules, regulations, procedures and manuals relevant to his duties, and found fit;
- (xxvii) “chief controller” means an authorised person in overall charge of OCC functions;
- (xxviii) “Coded Manual Mode or ATP mode” means the mode of operation of train under continuous automatic train control system where train is driven manually but remains subject to maximum speed determined by automatic train protection system;
- (xxix) “Commissioner” means the Commissioner of Metro Railway Safety appointed under section 7 of the Act;
- (xxx) “Communication Based Train Control (CBTC) system is a continuous, automatic train control system utilizing high-resolution train location determination, independent of track circuits; continuous, high-capacity,

- bidirectional train-to-wayside data communications; and train-borne and wayside processors capable of implementing vital functions;
- (xxxi) “competent staff” means the person competent to execute the work allotted to him and possessing valid competency;
- (xxxii) “connections”, when used with reference to a running line, means the arrangements used to connect such line with other lines or to cross it;
- (xxxiii) “Continuous Automatic Train Control (CATC) system” means an automatic system of controlling and monitoring train movements continuously by means of sub-systems, namely, Automatic Train Protection System, Automatic Train Operation system and Automatic Train Supervision System;
- (xxxiv) “cross passage” means a physical connection between two single track tunnels, which is used for maintenance of the metro railway, and can also be used for evacuation of passengers and for other relief works during emergencies;
- (xxxv) “Cut-Out-Mode” means the mode of operation of trains under the CATC system when train borne automatic train protection equipment is cutout;
- (xxxvi) “depot controller” means an authorised person responsible for movements of rakes within the depot area including interchange of rakes between the depot and the main line;
- (xxxvii) “departure order indication” is an indication on a display panel in the Train Operator’s cab conveying authority to the Train Operator to proceed or start his train;
- (xxxviii) “Driverless Train Operation (DTO)” means a train operation where a Roving Attendant is on-board the train, but normally not in the driving cab, and his or her duties shall be defined as per special instructions;
- (xxxix) “electrical way and works” means the traction power installations including overhead equipment or third rail equipment and other connected works provided on the electrified sections of the metro railway;
- (xl) “emergency” means an occurrence where there is an imminent or a continuing risk of injury and damage or major disruption to the metro railway service;
- (xli) “emergency stop plunger” means the device, the operation of which causes the trains, within the station limits, running on automatic train protection, to come to a stop;
- (xlii) “Emergency Train Operator (ETO)” means an authorised person having valid certificate of competency for train operation, nominated by Traffic Controller to drive the train in an emergency;
- (xliii) “engineer’s possession” means a defined section of track under the sole control of an authorised person for a specific length of time;
- (xliv) “engineering train unit” means a train which is used for the maintenance and repairs of track side fixed infrastructure;
- (xlv) “facing and trailing points” means points which are facing or trailing in accordance with the direction a train or vehicle moves over them and the points are said to be facing points when by their operation, train approaching them can be directly diverted from the line upon which it is running;
- (xlvi) “feeding post” means a supply control post in overhead traction power system, where the incoming feeder lines from grid sub-station are terminated;
- (xlvii) “fixed signal” means a signal of station at a fixed location controlling the movement of trains and forming part of the signalling system;

- (xlviii) “fouling mark” means the mark at which the infringement of fixed standard dimensions occurs where two lines cross or join one another;
- (xlix) “headway” means the time interval between two successive trains;
- (l) “incident” means any occurrence which causes or has potential to cause delay or disruption to passenger services;
- (li) “inspection car” means a self-propelled vehicle, which is used for the inspection, maintenance and repairs of the equipment of the metro railway;
- (lii) “insulated third rail joint” means a special joint for the purpose of sectioning the third rail being powered from different traction sub-stations;
- (liii) “interlocking” means an arrangement of signals, points and other appliances, operated from a panel or work-station, so interconnected by mechanical or electrical or electronic locking, or any combination thereof, that their operation must take place in proper sequence to ensure safety;
- (liv) “irrevocable emergency brake” means an emergency brake when applied, intentionally or otherwise, remains applied until the train speed comes to zero and the cause of application of the emergency brake is removed or reset;
- (lv) “isolation” means an arrangement secured by the setting of points, or other approved means, to protect the line so isolated from obstruction due to movement on other connected line or lines;
- (lvi) “jumper cable” means a cable provided with clips for use as a temporary electric connection to bridge a gap in a running rail, cable or pipe;
- (lvii) “local control” means the assumption of the responsibilities of the Traffic Controller for a specific station by a person who is authorised to do so for the time being;
- (lviii) “metro railway employee” means an employee of the metro railway duly qualified, possessing a valid certificate of competency and nominated to undertake and perform the duties entrusted to him;
- (lix) “moving block” means the block system where the blocks are in real time defined by signalling and train control system as safe zones around each train.
- (lx) “neutral section” means a short section of insulated and dead overhead equipment, which separates the areas fed by adjacent sub-station or feeding post for AC traction power system;
- (lxi) “normal direction of traffic” means traffic moving on the left side track;
- (lxii) “obstruction” with its cognate expressions includes a train, vehicle or obstacle on or fouling a line or any condition which is dangerous to trains;
- (lxiii) “Operations Control Centre (OCC)” means the organisation in overall charge of controlling the movement of trains on the main line and includes limited backup facilities, where provided which may be located at different locations to meet contingencies and termed as backup control centre;
- (lxiv) “overhead equipment” means the electrical conductors over the track together with their associated fittings, insulators and other attachments by means of which they are suspended and registered in position for the purpose of electrical traction;
- (lxv) “parted”, when used with the reference of train divided, means the train is physically divided into two parts and is not coupled;
- (lxvi) “passenger train” means a train intended for the movement of passengers and their baggage;

- (lxvii) "Permanent Speed Restriction (PSR)" means the speed restriction which is perpetually imposed and incorporated through CATC system or as per special instruction;
- (lxviii) "permission to approach" means permission given for a train to enter the block section;
- (lxix) "Platform Screen Doors" means a system of automated doors synchronized with the train doors which are provided at the platform edge to isolate passengers on platform from track;
- (lxx) "platform supervisor booth" means a monitoring cabin on platform for supervisor;
- (lxxi) "point and trap indicators" means appliances fitted to and working with points to indicate the position in which they are set;
- (lxxii) "power block" means withdrawing traction power supply from a particular section;
- (lxxiii) "proceed code" means the automatic train protection code other than zero speed code on the Train Operator's console which indicates the target speed;
- (lxxiv) "Railway Board" means the Railway Board as empowered under section 2 of the Indian Railway Board Act, 1905 (4 of 1905);
- (lxxv) "receiving sub-station" means an electric sub-station where electric power supply is received from grid sub-station and transformed to appropriate voltage for distribution;
- (lxxvi) "Restricted Manual (RM) Mode" means a driving mode where train is driven manually and is subjected to automatic train protection in respect of maximum speed limit only;
- (lxxvii) "rolling stock controller" means an authorised person responsible for the control of the rolling stock systems of the metro railway;
- (lxxviii) "rolling stock supervisor" means an authorised person duly qualified to examine trains and certify their fitness for safe running;
- (lxxix) "Roving Attendant" means an authorised person responsible for performing required functions in driverless or unattended train operation;
- (lxxx) "running line" means the track used for running trains through and between stations and includes connections, if any, used by a train when entering or leaving stations;
- (lxxxi) "running train" means a train which has started on authority to proceed but has not completed its journey;
- (lxxxii) "Run-on-Sight (ROS) Mode" means a driving mode where the train is driven manually and is subject to ATP restriction in respect of speed only until ATP track indications are recognised after which it automatically changes to coded manual mode;
- (lxxxiii) "SCADA" means supervisory control and data acquisition system for the purpose of remote monitoring and control of all traction power supply, installations and auxiliary systems;
- (lxxxiv) "schedule" means a schedule annexed to these rules;
- (lxxxv) "secure a train" means no motoring command is given and brake is applied so as to ensure no movement of the train;
- (lxxxvi) "short circuiting device" means such devices provided for safety reasons on each station located in auxiliary sub-station or Traction sub-station or station

- control room to temporarily short circuit the running rails to earth in case the rise of running rail potential exceeds the prescribed limits;
- (lxxxvii) “shunting” means the movement of a coach or coaches with or without traction motor or of any other self-propelled vehicle, for the purpose of attaching, detaching or transfer or for any other purpose;
- (lxxxviii) “signal” means an indication given to a Train Operator for controlling the movement of the train;
- (lxxxix) “signal supervisor” means an authorised person to examine signalling system and certify signalling system fitness for the safe running of trains;
- (xc) “signalling system controller” means an authorised person responsible for the control of signalling system maintenance and coordination with other controllers;
- (xci) “sleep mode” is the state where all systems of the train are switched off except the sub-system which initiates the wake-up test;
- (xcii) “sleep process”, means the process initiated by UTMS either manually or automatically to switch off all systems of the train except the sub-system which initiates the wake-up;
- (xciii) “special instruction” means instruction issued from time to time by the authorised officer in respect of particular cases or special circumstances;
- (xciv) “staff protection key” means a key used by an authorised person for going to a particular track section for inspection, maintenance, cleaning or any other work;
- (xcv) “staff special key” means a key provided to open Manual Secondary Door from platform side which is used by the maintenance staff to access the track side;
- (xcvi) “station” means any place on a line of the metro railway at which passenger traffic is dealt with;
- (xcvii) “Station Controller” means the person on duty who is for the time being responsible for the working of the station and traffic within station limits and includes the Assistant Station Controller or any person who is for the time being in independent charge of the working of such station and traffic;
- (xcviii) “station control room” means the room where station control panel or work-station is located;
- (xcix) “station limits” means the limits or chainages as defined in the station working orders;
- (c) “supply control post” means an assembly of interrupters, isolator switches, remote control equipment and other apparatus provided for controlling power supply to overhead equipment or third rail traction equipment, and it includes feeding posts, sectioning and paralleling posts, and sub-sectioning posts in the case of overhead equipment;
- (ci) “system of working” means one or more of the systems specified in Chapter VIII for the time being for the working of trains;
- (cii) “target distance” means the farthest point to which the train may safely proceed;
- (ciii) “target speed” means the speed displayed on the Train Operator’s console which the train must not exceed;
- (civ) “telecom system controller” means an authorised person responsible for the control of telecommunication systems;
- (cv) “temporary earth” means an additional earthing device which is applied after the issue of a ‘permit-to-work’ and removed prior to the cancellation of the ‘permit-to-work’;

- (cvi) “Temporary Speed Restriction (TSR)” means the speed restriction, in no case greater than PSR, imposed by an authorized person to limit speed of trains over a section of track in accordance with special instructions.
- (cvii) “terminal station” means the station at the end of a line;
- (cviii) “test track” means the portion of the track in the depot used for testing of a train;
- (cix) “third rail” means a rail on insulators laid by the side of running rails for conduction of electric current to the train through current collector fitted on the train for its running on the main line and depots in a DC traction system;
- (cx) “track and structure or works engineer” means an authorised person responsible for the construction or maintenance of points, underground structure, surface structure, bridges or other works connected therewith;
- (cxi) “track circuit” means an electrical circuit provided to detect the presence or absence of a vehicle on a portion of track, the rails of the track forming part of the circuit;
- (cxii) “traction”, when used with the reference of train, means the act of drawing or pulling of a vehicle along a surface by motor power;
- (cxiii) “traction power” means a traction power system working on 25000 volts single phase, 50Hz alternating current or 750 volt or some other voltage of Direct Current with Overhead Equipment System, or on 750 volts or some other voltage of direct current with third rail system;
- (cxiv) “traction feeder breaker” means a circuit breaker controlling the traction power supply to the third rail or overhead equipment;
- (cxv) “traction power controller” means an authorised person responsible for the control of the traction and auxiliary power distribution systems of the metro railway;
- (cxvi) “Traffic Controller” means an authorised person on duty in the OCC who is responsible for running of trains on a section of metro railway;
- (cxvii) “traffic hours” means the period between the time of the start of the running of the first scheduled train and termination of the last scheduled train;
- (cxviii) “train” means an engine with or without vehicle attached or self-propelled vehicle with or without a trailer which cannot be readily lifted off the track;
- (cxix) “Train Control and Monitoring system (TCMS)” or “Train Integrated Management System (TIMS)” means a system designed to provide information and exercise control on a variety of functions related to movement of metro trains, like traction, power, braking, air conditioning, etc;
- (cxx) “train divided”, means that the train integrity is lost and train may be incomplete or not coupled or electrically uncoupled;
- (cxxi) “train integrity”, means the train is complete and coupled;
- (cxxii) “Train Operator” means the driver of the train responsible for movement and control of the train;
- (cxxiii) “train radio” means a wireless telephone message communication system between the cab of the train, stations and the OCC;
- (cxxiv) “tunnel ventilation section” means a minimum length of a tunnel section of an up or down line capable of being mechanically ventilated using tunnel ventilation fans;
- (cxxv) “Unattended Train Operation (UTO)” means a train operation with no regulatory requirement of Roving Attendant on-board the train and emergency

handling or abnormal situation shall be handled by system or through special instruction;

- (cxxvi) “Unattended Train Management System (UTMS)” means metro rail transportation system with self-propelled vehicle and operated on the guided way, which monitors and controls, starting and stopping of the trains, operation of train doors and platform screen doors, without any regulatory requirement of the staff present in the train and detection and management of emergency situations as provided by system and/or staff in OCC including a Roving Attendant and station staff;
 - (cxxvii) “virtual signal” is an intermediate entry or exit location as displayed on interlocking and automatic train supervision monitors which may be used where required for dividing routes between two fixed signals and not a physical signal and only controls trains with automatic train protection;
 - (cxxviii) “wake-up-test” means the test performed by UTMS to ensure that required functions of the train are working correctly;
 - (cxxix) “wake-up process”, means process initiated by UTMS either manually or automatically to switch on all systems of the train and to initiate the wake-up;
 - (cxxx) “work train” means a departmental train intended solely for execution of works, including maintenance works on the metro railway network.
- (2) Words and expressions used in these rules and not defined but defined in the Act, the Metro Railways (Operation and Maintenance) Act, 2002 (60 of 2002), or the Metro Railways (Construction of Works) Act, 1978 (33 of 1978), shall have the meanings respectively assigned to them in those Acts.

CHAPTER II

RULES APPLYING TO AUTHORISED PERSONS GENERALLY

3. **Supply of copies of rules** —The metro railway administration shall supply a copy of these rules and amendments made therein to—
 - (a)
 - (i) operation control centre;
 - (ii) each station;
 - (iii) each rake maintenance depot, traction depot, permanent way depot and signal depot; and
 - (iv) such other offices as may be specified under special instructions; and
 - (b) each authorised person on whom responsibility has been placed under these rules, or of such portions of rules as relate to his duties.
4. **Upkeep of the copy of the rules** —Every authorised person who has been supplied with a copy of these rules shall, —
 - (a) keep it posted with all corrections;
 - (b) produce the same on demand by any of his superiors;
 - (c) obtain a new copy from his superior in case his copy is lost or defaced; and
 - (d) ensure that the staff working under him are supplied with all corrections or amendments and that they comply with the provisions of these rules.
5. **Knowledge of rules and issue of certificate of competency** —
 - (1) Knowledge of rules —Every authorised person shall —
 - (a) be fully conversant with the rules relating to his duties;
 - (b) pass the examinations as specified by the authorised officer of the metro railway administration;

- (c) satisfy himself that the staff working under him are conversant with the rules relating to their duties and obtain a written assurance from them.
- (2) Issue of certificate of competency —No authorised person shall be assigned any duty under these rules unless he has passed the prescribed examination as regards his technical ability and skills and knowledge of rules, regulations, procedures and manuals relevant to his duties and has been issued, after being found fit, a certificate of competency by an officer specifically nominated by the authorised officer, in this regard.
- 6. **Assistance in observance of rules** — Every authorised person shall render assistance in carrying out these rules and report promptly any breach thereof, which may come to his notice, to his superior officer and other authority concerned.
- 7. **Obedience to rules and orders** —Every authorised person shall observe and obey—
 - (a) all rules and special instructions; and
 - (b) all lawful orders given by his superior officials.
- 8. **Prevention of trespass, damage or loss** —
 - (1) Every authorised person shall be responsible for the security and protection of the property of metro railway under his charge or possession.
 - (2) Every authorised person shall endeavour to prevent—
 - (a) trespass on metro railway premises;
 - (b) theft, damage or loss of metro railway property;
 - (c) injury to passengers, others and himself; and
 - (d) fire and other unsafe incidents in metro railway premises.
- 9. **Attendance to duty** —Every authorised person shall be on duty at such times and places and for such period as may be fixed by the metro railway administration and shall also attend at any other time and place where his services may be required.
- 10. **Absence from duty** —
 - (1) no authorised person shall, without prior permission of his superior officer, absent himself from duty or alter his appointed hours of attendance or exchange duty with any other authorised person or leave his charges of duty unless properly relieved;
 - (2) in case any authorised person, while on duty, desires to absent himself from duty on the ground of illness, he shall immediately report the matter to his superior officer and shall not leave his duty unless an authorised person has been placed incharge thereof.
- 11. **Taking alcoholic drink, sedative, narcotic, stimulant drug or preparation** —
 - (1) An authorised person shall not take or use any alcoholic drink, sedative, narcotic or stimulant drug or preparation within eight hours before the commencement of his duty or take or use any such drink, drug or preparation while on duty.
 - (2) No authorised person, while on duty, shall be in a state of intoxication or in a state in which, by reason of his having taken or used any alcoholic drink, sedative, narcotic or stimulant drug or preparation, his capacity to perform his duties is impaired.
 - (3) An authorised person while on duty, shall not smoke or chew tobacco.
- 12. **Conduct of authorised persons** —
Every authorised person shall, —
 - (a) wear the uniform and badge as prescribed by the metro railway administration and be neat and tidy in his appearance while on duty;
 - (b) be prompt, civil and courteous;
 - (c) not solicit or accept illegal gratification;

- (d) give all reasonable assistance and be careful to give correct information to the public;
- (e) make complete and truthful statement at all times in all reports pertaining to his duty; and
- (f) when asked, give his name and designation without hesitation.

13. Duty for ensuring safety —

- (1) Every authorised person shall—
 - (a) see that every effort is made for ensuring safety of the public and of his fellow person;
 - (b) promptly, report to his superior any occurrence likely to affect the safe and proper working of the metro railway which may come to his notice; and
 - (c) render spontaneously all possible assistance when called upon to do so by the appropriate official in case of an accident or obstruction.
- (2) Every authorised person who observes —
 - (a) anything wrong with a train; or
 - (b) any obstruction, failure or threatened failure of any part of the way or works or overhead electric equipment or third rail electric equipment including power supply installation; or
 - (c) any defective signal; or
 - (d) any unusual circumstances such as fire, smoke, flood accident or other dangerous condition on any part of the system likely to interfere with the safe running of trains, or the safety of the public, shall take immediate steps, to prevent the accident and promptly report the matter to the OCC or the nearest Station Controller.

14. Standard time—The working of trains between stations on metro railway shall be regulated by the Indian Standard Time.

15. Access control—

- (1) All authorised persons, in addition to identity cards, shall be provided with access control cards or authorisation signed by their controlling officers to allow them to visit places of metro railway system with restricted access in line of their duties.
- (2) The following places in particular shall have restricted access in addition to other places as notified from time to time by the order of authorised officer, namely: -
 - (a) guideways, viaducts and tunnels;
 - (b) receiving sub-station;
 - (c) traction substation;
 - (d) auxiliary substation;
 - (e) signalling equipment room;
 - (f) telecommunication equipment room;
 - (g) uninterrupted power supply room;
 - (h) station control room;
 - (i) operation control centre;
 - (j) back-up control centre, where provided;
 - (k) depot control centre;
 - (l) Demarcated UTO territory in Depot.

CHAPTER III

SIGNALS AND CONTROL

16. General —

- (1) The following signals shall be used for controlling the movements of trains on the metro railway, namely: -
 - (a) cab signals;
 - (b) fixed signals and equipments;
 - (c) hand signals;
 - (d) Virtual signals.
- (2) The aspects displayed by fixed signals are the same by day and the night, in open and in tunnels.
- (3) A fixed signal shall be placed where practicable, on the left hand side of the track to which it refers, unless authorised otherwise under special instructions and shall be visible from such a distance as shall enable a Train Operator to brake a train from twenty-five kilometer per hour speed to stop before reaching the fixed signal and a repeater signal shall be provided at locations where due to obstructions such visibility is not available, except in depot: Provided that where high speed cut out mode is provided, this speed shall be forty kilometer per hour.
- (4) The signal sighting distance as specified in sub-rule (3) shall be specified under special instructions.
- (5) Axle counters or track circuits, where provided, to assist operation of non-ATP equipped trains or trains with inoperative on-board ATP equipment.

17. Description of signals —

(1) Cab Signals -

- (i) Train movements on running tracks and in depot (if equipped with ATP) shall normally be governed by the automatic train protection system which displays to the Train Operator in the operating console, -
 - (a) actual speed of the train;
 - (b) the maximum permitted speed at each point of travel;
 - (c) the distance the train is currently authorised to travel (where provided);
 - (d) system alarms; and
 - (e) messages.
- (ii) in case the target speed indication and the target distance indication, where provided, are greater than zero, the indication is referred to as “PROCEED” indication;
- (iii) in case either of these indications is “0”, the indication is referred to as “STOP” indication;
- (iv) the Train Operator is authorised to drive his train upto the indicated speed as far as authority has been given for such purpose.

(2) Fixed Signals—

- (i) on main lines, fixed signals are colour light signals, either showing two aspects or three aspects;
- (ii) two aspects fixed signals, where provided shall be capable of showing a “red” or a “white or violet” aspect, as under, -

- (a) a “red” aspect indicates that a train must be brought to Stop short of the signal when driven in RM or ROS or cut out mode;
 - (b) a “white or violet” aspect indicates that the route is set and locked but may not be fully clear upto the next fixed signal and a train operating under cab signals may proceed under the authority of the cab signals but a train operating on the sole authority of line side signals must stop and seek instructions from the Traffic Controller;
 - (iii) three aspect signals, where provided shall be capable of showing a “red”, “white or violet” or “green” aspect, as under, -
 - (a) a “red” aspect indicates that a train must be brought to Stop short of the signal when driven in RM or ROS or cut out mode;
 - (b) a “white or violet” aspect indicates that the route is set and locked but may not be fully clear upto the next fixed signal and a train operating under cab signals may proceed under the authority of the cab signals but a train operating on the sole authority of line side signals must stop and seek instructions from the Traffic Controller;
 - (c) a “green” aspect indicates that the route is cleared to the next fixed signal and the train may proceed as far as the next fixed signal;
 - (iv) when a fixed signal is not in use, the aspect shall be covered and the cover shall display two crossed white bars on a black background, the bars being not less than thirty centimetres long and ten centimetres wide;
 - (v) fixed signals on main line may be kept blank when cab signals are available under ATP or ATO or DTO or UTO mode of operation and should be lit for trains under RM or ROS or cut-out mode of operation.
- (3) Fixed Signals in Depots —
- (i) in depots, fixed signals may be colour light or position light type. Colour light type using red and yellow aspects should be used in the manner as specified below, -
 - (a) a “red” aspect indicates that a train must be brought to stop short of the signal when driven in RM or ROS or cut out mode;
 - (b) a “yellow” aspect indicates that the route is set, locked and clear and a train may proceed as far as the line is clear and the Train Operator must keep a look out for any obstruction;
 - (ii) in position light type signalling, -
 - (a) two white lights displayed horizontally shall mean that a train must stop;
 - (b) two white lights displayed at an angle shall mean that a train may proceed as far as the line is clear and the Train Operator must keep a lookout for the obstruction;
 - (iii) the depots and stabling lines shall be isolated from the running line through approved means.
- (4) Hand Signals —
- (i) hand signals shall normally be used only for the shunting of work train in depotor at the site of work or in extreme emergency;
 - (ii) any light other than “green” or any object waived violently shall be interpreted as a stop signal;
 - (iii) “STOP” shall be indicated by, -
 - (a) a red lamp;
 - (b) raising of both arms above the head;

- (c) waving a white light rapidly from side to side;
- (d) a red flag;
- (iv) “PROCEED” shall be indicated by a green lamp or green flag held steadily;
- (v) hand signals for shunting and train movements shall have the following configurations:-

S. N.	ASPECT	INDICATION
(a)	A green lamp or green flag moved slowly up and down	Move away from the signal
(b)	A green lamp or green flag waved from side to side across the body	Move towards the signal
(c)	Display red light or red flag	Stop

- (vi) when during cautious driving or shunting, the speed of a train is to be reduced, the hand signal for movement shall be given at a slower and slower rate and when a stop is required, a STOP signal shall be given;
 - (vii) each station control room shall have at least one hand lamp capable of displaying red, green and white aspects, one red and one green flag readily accessible and in working order and each Station Controller shall be conversant with their location and their proper use;
 - (viii) each authorised person involved in the shunting of work trains, the operation of work trains at a maintenance site and the operation of work trains within an engineer’s possession, shall carry a hand lamp capable of displaying red, green and white aspects in working order and shall be conversant with its proper use and in addition, he shall also carry a red flag and a green flag.
- (5) Virtual signal—

These are not physical signals but locations on line. These are used as an intermediate entry or exit location or as name given, in the yard or line layout and displayed on interlocking and ATS monitors. These may be used where required for dividing routes between two fixed signals. This signal controls trains with ATP to enable shorter movements.

18. Provision of signals —

- (1) Fixed signals shall be provided on running lines at the approach to all points and crossings of interlocked areas and located in such a way that trains stop at a safe distance from any fouling movement or location.
- (2) All depot tracks and any other tracks not equipped with ATP shall be controlled by fixed signals for both entry to and exit from the main line.
- (3) All single ended tracks shall be provided with at least one permanent red aspect light to indicate the point beyond which train shall not proceed.
- (4) Stopping markers shall be provided for all platforms to indicate where a train of a given length shall stop for the convenient detraining and entraining of passengers:

Provided that where trains of varying lengths operate, separate markers shall be provided for trains of each potential length and for bi-directional running.

19. Working of signals and points —

- (1) Control of signals and points shall be from a route setting panel or work-station, and complete routes, points and signals, shall be cleared by a single set of operations under normal conditions.

- (2) Any failure of vital equipments shall cause the signalling system to display the most restrictive indication.
- (3) Any route which has been cleared for a train shall not be cancelled until it is cleared by the train entering the route except: —
 - (a) in case of emergency; and
 - (b) in case where operating conditions require that an alternate route be cleared and in this case, the alternative route shall not be made available for clearance until the pre-set time, defined under special instructions, has elapsed from the time the original route was cancelled.
- (4) In conditions of failure of route setting controls, points can be set individually from OCC, station control room or locally, as may be necessary.

20. Control of signalling —

- (1)
 - (i) Main line signalling is controlled from the OCC and operates normally under automatic control with routes being set and train intervals regulated by computer control;
 - (ii) signalling system shall permit more than one train at a time in one tunnel ventilation section. However signalling system can restrict one train in one tunnel ventilation section as per special instructions. The Traffic Controller can also restrict to one train only in one tunnel ventilation section if required;
 - (iii) the Traffic Controller, if required, may hand over control of the signals of a specific station to the Station Controller and a local control panel or work-station shall be provided at the station for this purpose;
 - (iv) in case of complete failure of control from the OCC, the entire control may be transferred to a standby backup control centre, if provided at an alternate location, which may be at separate locations for traffic control and control of traction power;
 - (v) the Traffic Controller shall have control of all routes on and to the main line as specified in special instructions;
 - (vi) train movement shall normally be under computer control but the Traffic Controller has the capability of setting routes manually and of setting at individual points, if necessary.
- (2) Safety Communication —
 - (i) all communication between the OCC and Train Operators, Station Controller, maintenance staff of electrical, rolling stock, signal and telecommunication and track structures department and other pertaining to movement of trains shall be recorded with time stamping, and preserved for incident analysis and training and the mode of preservations and its duration shall be as specified in special instructions;
 - (ii) the authorised person shall initiate and acknowledge radio messages in a manner that ensures establishment of communication only between intended parties;
 - (iii) messages affecting train movements shall be addressed by the Traffic Controller to only one train at a time:

Provided that in an emergency, a blanket message may be sent by the Traffic Controller to all trains in or approaching a particular area, which must be acknowledged individually by all concerned Train Operators:

Provided further that in an emergency, such message may be given by the Station Controller, if authorised by the Traffic Controller in this behalf.

(3) Running Lines —

- (i) the signalling system on the running line shall be an ATP system.
- (ii) in case the speed which permits a train running on the ATP to stop under normal braking within a limit of safety is exceeded, and the Train Operator fails to take corrective action on the alarm generated, an irrevocable emergency brake shall automatically apply which ensures that the train does not proceed beyond the safe limit;
- (iii) normal operation of the train running on the ATP is monitored from indication on the Train Operator's console and each main running line is duly signalled for operation;
- (iv) certain lines are equipped for ATO with provision for the Train Operator to assume manual control of the train if required;
- (v) in the event of failure of the automatic train operation equipment and on lines not so equipped, Train Operators shall control their train upto the speed indicated on the console;
- (vi) where no cab signal is available, but target and current running speed from on board ATP system are available or displayed on TCMS or TMS, the Train Operator shall proceed under the ATP not exceeding the speed shown on the TCMS or TMS else movement of train shall be restricted to a maximum speed of twenty-five kilometres per hour in non ATP or as per special instructions;
- (vii) fixed signals are to be provided at approach to all points and crossings of the interlocked area and certain other locations.

(4)

- (i) Station Controller's control panel or work-station shall have the capability of setting routes within the area of control and of setting individual points when necessary;
- (ii) on each platform and station control room, there shall be emergency stop plunger, which when operated shall cause any approaching train to stop before it enters the platform. In this situation any train entering, stationary or leaving the platform on ATP shall experience a full irrevocable emergency brake application.

(5)

- (i) The depot tracks may not be equipped with full complement of ATP and trains shall be automatically restricted to a maximum speed of twenty five kilometres per hour and the Train Operator may further restrict the speed of the train in the depot lines to a lower value as specified in special instructions;
- (ii) movements within depots shall be controlled by fixed colour light signals or position light type signals;
- (iii) control of movements within the depot shall be exercised by a depot controller from a control panel or work-station located within the depot or at some other suitable location;
- (iv) the depot controller shall use a control panel or work-station for setting and clearing of routes within the depot;
- (v) instructions to Train Operators as to which siding they shall drive or give permission to depart may be given by a train radio on a dedicated depot channel distinct from that used by the OCC;
- (vi) a verbal instruction, except in UTO or DTO or ATP mode and proceed indication from the fixed signals, where provided shall be received before a train moves into a depot from the main-line;

- (vii) both verbal instruction, except in UTO or DTO or ATP mode and proceed indication from the fixed signals, where provided, shall be received before a train moves into mainline from a depot.

(6)

- (i) Local operation of points on running lines as and when required may only be undertaken by a Station Controller with the permission of the Traffic Controller. Such transfer of control should be acknowledged by the Station Controller before the local control can be activated;
- (ii) the crank handle used to manually operate the points is interlocked through a key which, when released by the Station Controller, shall inhibit setting of routes over the points concerned;
- (iii) some points in depots may be trailable. In the event of a route failure they need to be correctly set before a movement is authorised over them in the trailing direction. The non-trailable points shall be protected by fixed signals or stop board. Backing of trains over trailable points is strictly prohibited except under special instructions.

(7)

- (i) Where some lines of the network are equipped with the ATO, the Train Operator shall close the train doors and start the train from a station and monitor its functioning;
- (ii) all passenger trains shall be equipped with ATP equipment and can be driven manually under the control of the ATP system.
- (iii) in case ATP information is not available from the track side equipment, the train can be driven with Traffic Controller's permission in "Run On Sight" mode, or in "Restricted Manual" Mode, where the train is not equipped with Run On Sight mode, in which the on board ATP equipment restricts the speed of the train to twenty-five kilometer per hour and when the ATP information is again available from the track, the mode automatically upgrades to coded manual;
- (iv) in case the ATP is not available from the track or loco or cab side equipment due to fault or otherwise, the train may be driven with Traffic Controller's permission in 'Restricted Manual' mode and in this mode, the on board ATP equipment restricts the speed of the train to twenty-five Kilometres per hour, this mode shall be used in depots and exceptionally on running lines when authorised by the Traffic Controller;
- (v) in case the on board ATP equipment is defective, the train may be driven in high speed cut out mode, where available, or low speed cut out mode, as authorised by the Traffic Controller and when so authorised, the Train Operator may open and operate a sealed switch and drive the train without ATP and in this mode the Train Operator shall limit the speed of the train to forty kilometres per hour in high speed cut out mode, where available, and twenty five kilometres per hour in low speed cut out mode and also be responsible for safe movement of the train.

21. Train detained on Line —

- (1) In case a train operating under the ATO or cab signals comes to a stop on a running line and does not receive a proceed code within sixty seconds, the Train Operator shall inform the Traffic Controller by the radio and ask for instructions.
- (2) The Traffic Controller shall check his indications and if he is satisfied that there is no train ahead, he may instruct the Train Operator to proceed in ROS Mode or RM Mode, as the case may be, exercising utmost vigilance so that he may stop short of any obstruction.
- (3) In case, after travelling some distance, a proceed indication is received by the train, the train shall automatically upgrade to the Coded Manual Mode of control and the Train

Operator shall again inform the Traffic Controller. If the train is fitted with ATO, the AM of the control may be selected.

- (4) In case the cause of the problem is confirmed as a track side equipment malfunction, the Traffic Controller may instruct subsequent trains without waiting for the Train Operator to seek instructions, provided the preceding train has passed through the affected section and reported resumption of Coded Manual Mode.
- (5) A train operating without receipt of proceed code shall stop at fixed signals irrespective of their aspect, and the Train Operator shall contact the Traffic Controller by the radio and ask for instructions.
- (6) In case a train operating under the authority of line side signals only, stops at a fixed signal at danger and it does not clear within sixty seconds, the Train Operator shall inform to the Traffic Controller by radio and ask for instructions.
- (7) The Traffic Controller shall consult the concerned Station Controller in control of the signalling and decide whether or not the train may proceed.
- (8) In case there are no points in the route ahead, the Traffic Controller may instruct the Train Operator to proceed as far as the next fixed signal at a reduced speed, not more than twenty-five kilometres per hour, such that he can stop short of any obstructions and if after travelling some distance, a proceed indication is received by the train running on RM Mode, or ROS Mode, the train shall automatically upgrade to the Coded Manual Mode of control, where provided, and the Train Operator shall again inform the Traffic Controller; and if the train is fitted with ATO, AM of the control may be selected.
- (9) In case there are points in the route, the Traffic Controller and the Station Controller shall examine the indication of the points and if indications at the station and the OCC agree that all points are set correctly and locked, the Traffic Controller may instruct the Train Operator to proceed as far as the next fixed signal at a reduced speed, such that he can stop short of any obstruction and if after travelling some distance, a proceed indication is received by the train running on RM Mode or ROS Mode, as applicable, the train shall automatically upgrade to the Coded Manual Mode of control, where provided, and the Train Operator shall again inform the Traffic Controller; and if the train is fitted with ATO, AM of the control may be selected.
- (10) In case points are not indicated as set and locked, the Traffic Controller shall instruct the Station Controller to examine the concerned points. If they are lying in the incorrect position, he shall manually set the points from the station control room with the crank handle interlocked through a key which, when released, shall inhibit setting of routes over the points concerned, and the points shall be secured with a clamp and padlocked and the train instructed by hand signal to proceed.

22. Absence of cab signalling —

- (1) Failure of the ATP shall initiate an irrevocable emergency brake application.
- (2) The Train Operator shall report the occurrence to the Traffic Controller and seek instructions.
- (3) The Traffic Controller shall verify as far as practicable from indications on his control panel or work station that the problem is not caused by a track side fault or another train and if no cause is apparent, he shall instruct the Train Operator to select the RM Mode or ROS Mode and try to move ahead exercising utmost vigilance so that he may stop short of any obstruction.
- (4) In case the train still does not move, the Traffic Controller shall authorise the Train Operator to operate the cut out switch and to proceed in cut out mode of control upto a suitable station, so as to detrain the passengers and withdraw the train as per special instructions. During the cut out mode of control, the Train Operator shall exercise utmost vigilance so that he is able to stop short of any obstruction.

23. Failure of fixed signals —

- (1) In case a Train operator observes that a fixed signal is displaying:
 - a) no aspect (is blank); or
 - b) multiple aspects; or
 - c) flickering of aspects,then the signal should be treated as ON (Red) in case train is not running under the authority of cab signals and Train Operator shall stop his train and seek instructions from Traffic Controller by radio. However, trains running under the authority of cab signal may still pass the signal.
- (2) In case a train not running under the authority of cab signals stops at signal under sub-rule (1), then all concerned staff shall follow provisions of sub-rules (6) to (10) of rule 21 and sub-rules (1) to (4) of rule 22.

24. Failure of route setting —

- (1) In case a route through an interlocked area cannot be set automatically or by manual control from the OCC, the control of the area shall be transferred to the local station control room.
- (2) In case the route cannot be set, the Traffic Controller shall instruct the Train Operator to secure his train and wait for instructions.
- (3) The Traffic Controller and the Station Controller shall examine the indications for the points and if indications at the station and at the OCC agree that all points are set and locked correctly and blocked as per special instructions, the Traffic Controller may instruct the Train Operator to select ROS mode or RM Mode, as the case may be, and proceed at reduced speed such that he can stop short of any obstruction. Once cab signals show a proceed indication, Coded Manual Mode shall be resumed; and if the train is equipped with ATO, then AM may be resumed.
- (4) In case any point indication is missing or shows the points set for the wrong direction, or there is a discrepancy between the indications at the station and in OCC, the Station Controller shall examine the position of the points himself. If they are lying in the wrong direction, he shall manually set the points with the crank handle from the station control room and the points shall be secured with a clamp and padlock and the train instructed by hand signal to proceed and if after travelling some distance, a proceed indication is received by the train running on RM Mode or ROS Mode, the train shall automatically upgrade to the Coded Manual Mode of control, where provided and the Train Operator shall inform the Traffic Controller. If the train is fitted with ATO, AM of the control may be resumed.
- (5) The Station Controller, after examining the position of the points under sub-rule (4), shall then return to the station and report to the Traffic Controller that the route remains secured and any subsequent train may be instructed by radio to proceed.
- (6) In the depot, the depot controller may instruct the Train Operator by radio to proceed if—
 - (a) point indications for all points in the route show the points as set and locked in the correct position;
 - (b) In case points, for which indications are not available, have to be traversed, the depot controller shall ensure that the points are set manually and secured in correct position with the clamp and padlock before authorising the train to proceed.

CHAPTER IV
SPEED AND WORKING OF TRAINS

25. General —

- (1) No person shall drive a train unless he is in possession of a valid certificate of competency and medical fitness.
- (2) No Train Operator shall be booked to work a train until he has learnt the road and signed a certificate that he is fully acquainted with it and for this purpose, he shall be booked for minimum three round trips including one trip during night before being put to work the train independently.
3. (i) A Train Operator who has not worked on a section for three months or more should be given road learning trips to refresh his knowledge as under:

DURATION OF ABSENCE	NUMBER OF ROAD LEARNING TRIPS
3-6 months	1 round trip
Over 6 months	3 round trips

- (ii) An ETO who has not worked on a section for two years or more should be given road learning trips to refresh his knowledge as under:

DURATION OF ABSENCE	NUMBER OF ROAD LEARNING TRIPS
2-3 years	1 round trip
Over 3 years	3 round trips

- (4) No train shall be driven on a running line from the rear cab except —
 - (a) a locomotive working within an engineer's possession under the control of hand signals;
 - (b) in exceptional circumstances, when authorised by an official not below the grade of Traffic Controller, a look out shall be positioned in such cases at the leading end with the capacity to apply the emergency brake, and the speed of the train shall not exceed ten kilometre per hour in such cases.
- (5) In the depot, the train shall always be driven from the leading cab in the direction of travel or otherwise the Train Operator in the rear cab shall be instructed by cab to cab telephone by a second qualified Train Operator in the leading end cab.
- (6) In the depot, where a part consist or damaged train cannot be driven from the leading end, a look out shall be posted at the leading end, the Train Operator shall have the means of sounding an audible warning and, if practicable, the means of applying the emergency brake.
- (7) Each train while manned shall show two white lights to the front and two red lights to the rear in the direction of travel.
- (8) A stationary train on a running line shall be secured and shall show two red lights at each end of the train.
- (9) A train stabled in a depot or siding shall show at least one red light at each end at a double ended siding and at the outermost end in the dead end siding.

26. Service Regularity —

- (1) Every effort shall be made by OCC Staff, station staff and train staff to ensure that scheduled intervals between trains are maintained.
- (2) the Traffic Controller shall be responsible for maintaining the services at the scheduled level as far as practicable and for restoring the train services following a delay or disruption;
- (3) the Traffic Controller at the OCC may adjust the timetable and may intervene manually to set and clear routes if the timetable needs to be varied or in the event of a major disruption;
- (4) each Train Operator shall start his train from terminal station as soon as he gets the departure order indication.
- (5) the Train Operator shall start his train from intermediate station at the time indicated by the departure order indication.
- (6) the Train Operator shall follow any instruction from the Traffic Controller which varies the scheduled timings of his train.

27. Speed of trains —

- (1) Every train shall run on the line of the metro railway within the limits of speed as specified in the approved special instructions.
- (2) The speed of the trains in AM shall be automatically controlled by the CATC system.
- (3) In Coded Manual mode, the Train Operator shall regulate the speed of trains according to the cab signals, the maximum permissible speed shall not be exceeded and the speed of the train shall be such that it can be stopped within the distance indicated, as being the limit of safety, and the failure to do so shall result in irrevocable application of emergency brakes which shall be viewed as a failure on the part of the Train Operator.
- (4) In RM mode or ROS Mode, as the case may be, the maximum speed of the trains shall automatically be regulated to twenty-five kilometre per hour and the Train Operator shall further restrict the speed as required by special instructions.
- (5) In Cut out mode, the speed of train shall be manually regulated by the Train Operator as required and the power to the train propulsion system shall be cut off above twenty-five kilometre per hour in case of low speed cut out mode and, forty kilometre per hour in case of high speed cut out mode, where provided, and the Train Operator shall further restrict the speed as required by special instructions.
- (6) The maximum train speed when passing through a station platform shall be as per the approved special instructions, and in case, it is not possible to close any platform screen door or platform edge door, where provided, while receiving, dispatching or passing any train, the speed of the train shall be restricted to twenty kilometre per hour and an audible warning shall be sounded while entering or leaving the platform.
- (7) The maximum speed of trains in shunting shall not exceed the speed limit as prescribed in the approved special instructions in case of main line, and special instructions on other lines.
- (8) The speed of train in condition of poor visibility while operating in RM Mode or ROS Mode or Cut Out Mode shall be governed by special instructions.

- 28. Caution Order —**Whenever in consequence of the track or traction power equipment being under repair or for any other reason special precautions are necessary, a caution order detailing the kilometres between which such precautions are necessary, the reason for taking such precautions and the speed at which the train shall travel shall be handed over to the Train Operator at the stopping station short of the place where such precautions are necessary or at such other stations or work place and in such a manner as specified under special instructions and in addition, the Traffic Controller and the Station Controller shall ensure that the caution order for speed restriction has been suitably incorporated in the CATC system.

29. Train Staffing —

- (1) Each train shall be manned by only one Train Operator.
- (2) No person shall be allowed to travel in the Train Operator's cab except a cab authorised trainee or apprentice Train Operator, a driving inspector and other authorised person as laid down in special instructions.
- (3) Each Train Operator shall, at all times when on duty, be in possession of—
 - (a) a Train Operator's handbook containing these rules and the relevant operating and troubleshooting procedures for the train;
 - (b) two pairs of such spectacles he is required to wear under medical advice;
 - (c) a tri-colour torch or hand lamp capable of showing red, green and white aspects;
 - (d) first aid kit.
- (4) Each Train Operator when reporting for duty shall examine any notice issued for his guidance and in particular those which require special attention on the specific day and line.
- (5) The Train Operator shall undergo breathalyser test at the time of sign-on and sign-off.
- (6) The Train Operator shall check continuity and adequacy of brake power and air pressure or pneumatic pressure before taking the train out of depot and conduct brake power feel test at the first available opportunity.
- (7) In the event of a Train Operator becoming incapacitated while driving a train, he shall, if capable, inform the Traffic Controller who shall inform the Station Controller at the next station.
 - (i) In case the train is under ATO control, it shall arrive at the next station under automatic control and the Train Operator may be relieved at that station for medical attention.
 - (ii) The Station Controller, having valid competency certificate, may work as Train Operator till such time as another Train Operator is made available and the senior most authorised person qualified in the duties of the Station Controller, shall be in-charge of the station during the absence of the Station Controller.
 - (iii) In case the train is under manual control, it may get stopped between stations and in this case, the Station Controller at the station in rear of the train shall board the following train, Traffic Controller shall instruct the Train Operator to select RM mode or Cut out mode and proceed to the rear of the stalled train.
 - (iv) After reaching the rear of the stalled train, the Train Operator shall close down his driving position, secure the train and leave it in that condition until further instructions from the Traffic Controller.
 - (v) The Station Controller shall then board the stalled train, go to the leading cab and drive the train to the next station where the Train Operator may be relieved for medical attention and on arrival at the next station, the Traffic Controller shall be informed that the following train may now be authorised to resume the normal working.
 - (vi) The Station Controller shall continue to drive the train until such time when another Train Operator shall be made available. Once relieved by Train Operator, the Station Controller shall return to his station as speedily as practicable.
 - (vii) Alternatively, the Traffic Controller may arrange to provide a rescue Train Operator from the leading direction train, if necessary, as specified in special instructions.

30. Train defects —

- (1) No train with defective safety equipments, cab signalling, leading cab controls, interior lighting, ventilation, brakes or doors shall remain in passenger service, and it shall be removed from service at the earliest opportunity.

- (2) In case traction power is lost on any one bogie, the train need not be withdrawn from service, if power is lost on more than one bogie and the reduction in speed causes delay to following trains, the passengers shall be detrained at the station and the train worked empty, otherwise, the train may remain in passenger service until it can be replaced by a serviceable train.
- (3) In the event of a mechanical defect in the traction motor or drive which causes the axle to lock, the train shall be stopped immediately and shall not be moved until clearance has been given by a rolling stock supervisor through rolling stock controller.
- (4) In case a Train Operator discovers that the train cannot be driven or braked from the leading end cab, passengers shall be detrained as per the provisions referred in sub-rules (5) and (6).
- (5) Such an occurrence which may usually happen at a station and a terminal in which case the passengers shall be discharged at the station and dispatch the train to the depot, the Station Controller shall board the train and act as look out in the leading cab while the Train Operator drives from the rear cab in Cut Out mode. Information on signals and cleared routes shall be passed by cab to cab telephone and train speed restricted to a maximum of ten kilometres per hour.
- (6) In the unlikely event that the defect causes a train to stop between stations, the Traffic Controller shall arrange another Train Operator or if necessary a Station Controller competent to drive the train to reach the failed train and drive the train from rear cab in Cut Out mode with leading cab Train Operator acting as look out and the information on signals and cleared routes shall be passed by cab to cab telephone or radio and train speed restricted to a maximum of ten kilometres per hour. Alternatively, the failed train can be rescued by assisting train or shunting vehicle as per rule 50.
- (7) Upon failure of any train control system, which shall be indicated on the TCMS or TIMS, the Train Operator shall bring the train to a complete stop, if the train can still be operated by applying suitable isolations in the train control system, the Train Operator shall consult the Traffic Controller and seek further instructions, in consultation with rolling stock controller;
- (8) when such failure renders the train inoperative even by applying appropriate isolations in the TCMS or TIMS, the Train Operator shall inform the Traffic Controller that his train is stalled and requires assistance to move and in this case the train shall be dealt with in accordance with rule 50.
- (9) Failure of brakes to apply or to release shall be indicated on TCMS or TIMS and in case of failure of brakes to apply in one bogie, the train may be allowed to work with or without any speed restriction as per special instructions. Any such failure of brake to apply or to release shall be advised to Traffic Controller by the Train Operator. Whenever failure of brake to apply in terms of bogie count is such that one more bogie failure affects designed brake effort, the train shall be withdrawn at the end of the trip as per special instructions. However, in no case the train should be moved without release of all the brakes.
- (10)
 - (i) In case doors on a train are not closed, the train shall not start from a station;
 - (ii) in case doors cannot be closed by command or manually or there is no obviously open door and the "Doors Closed" indication is still not received, the train shall be worked as per special instructions;
 - (iii) in case some doors of train do not open at stations but after closing the doors, all doors are indicated as closed, the train may remain in passenger service, and necessary announcement shall be made in the train and on the stations to inform the passengers;

- (iv) in peak period, the extended dwell times at stations, that could result from some doors not opening, may make it desirable for the train to be withdrawn from service to avoid delaying the following trains.
- (11) Failure of an air conditioning unit shall be indicated on the TCMS or TIMS panel and although it has no effect on the safe operation of the train, the Train Operator shall report such failure to OCC, so that rake receives prompt attention to relieve hardship to passengers.
- (12)
 - (i) Failure of main car lighting on one or two cars shall be reported by the Train Operator to the OCC and the train may continue in passenger service to the end of its trip, provided the emergency lighting is working satisfactorily and at the terminal it shall be withdrawn from service or replaced by a good train;
 - (ii) in case all main train lightings fail or main lighting and emergency lighting both fail, on the same car, passengers shall be detained at the next station and the rake withdrawn from service.

31. Examination of trains —

- (1) Each train shall be examined by a competent person from the respective department before being offered for passenger service.
- (2) The examination shall ensure that all functions of the train are working correctly and in particular safety devices, such as:
 - (i) cab signalling;
 - (ii) safety brake circuits;
 - (iii) train radio communication;
 - (iv) head and tail lights;
 - (v) display panel of TCMS or TIMS,
 - (vi) brake gears;
 - (vii) isolating cock ties intact;
 - (viii) miniature circuit breaker and safety switch seals intact; and
 - (ix) any other item including the periodicity of item (vii) and (viii) above shall be as specified under special instructions.
- (3) The competent staff shall sign a certificate of safety test indicating duration of its validity, which shall remain in the leading cab in the direction of departure.
- (4) The Train Operator who runs the train from any stabling locations shall check that the certificate is upto date and currently valid before moving the train and he shall also check the head lights, the tail lights, the marker lights and the speedometer of the train.

32. Duties of a Train Operator —

- (1) The Train Operator shall pay immediate attention and obey every signal and shall always be vigilant and cautious and keep a sharp look out.
- (2) When a Train Operator, not working under Coded manual mode, approaches a fixed signal at 'on' or in 'defective' position, he shall not pass a fixed signal that refers to his train, unless after bringing his train to a stop, he is either given a written authority by the Station Controller to proceed past such signal or is authorised by the Traffic Controller on train radio in accordance with special instructions.
- (3) The Train Operator shall not operate the train at higher than the maximum authorised speed.
 - (4) The Train Operator shall be alert for changing rail conditions, and shall exercise extra care when operating in areas which may be affected by grease, oil, water or other substance, which could cause running rails to become slippery and shall adjust the train speed accordingly.

- (5) When closing the train doors, the Train Operator shall observe the platform end as far as practicable and prevent the closing doors from striking boarding passengers.
- (6) In case the doors are obstructed or the 'doors closed' indication is not received, the Train Operator shall reopen the doors and close them again.
- (7) In trains operating under AM, the Train Operator shall start the train after closing the doors and further progress to the next station and the opening of doors there takes place automatically, unless manual opening of doors is selected.
- (8) In trains not operating under AM, the Train Operator shall drive the train, observing and obeying cab signals, to the next station and shall stop the train at the appropriate stopping mark, and the doors of the train and platform screen doors, where provided shall not be opened until the train has come to a complete stop. The platform screen doors, where provided, can be opened simultaneously along with the train door open command or manually from the local panel provided near the cab on the platform.
- (9) In case a train stops short of its proper stopping place, even in AM, the Train Operator shall manually drive the train to its proper stopping place, and the doors of the train shall not be opened until the train has come to a complete stop and the platform screen doors where provided, can be opened manually from the local panel provided near the cab on the platform.
- (10) In case a train stops beyond its normal stopping place but with the doors still on the platform, the Train Operator shall contact the Traffic Controller to take his permission for reversing so as to align the train at proper stopping place and the doors may be opened manually and passengers allowed to alight and/or board.
- (11) In case the train stops beyond the end of the platform, the Train Operator shall seek instruction from the Traffic Controller and if the Traffic Controller can prevent the following train from approaching the platform by using the signalling controls, he may do so and then authorise the train at the platform to reverse until all doors are at the platform and train is correctly positioned at stopping place. Otherwise, an announcement shall be made to the passengers and the train may proceed to the next station without opening the doors, and if the train is the last train of the day, the train may be reversed on instructions of the Traffic Controller and stopped at the normal stopping place. If the train cannot be reversed, then the passengers may be allowed to disembark from the train by opening selected doors of the train by the exterior emergency doors control.
- (12)
 - (i) In case a passenger emergency alarm is operated in the train, the Train Operator shall try to establish voice communication with the location by intercom or public address and he shall try to establish the reason for the operation of alarm but, unless there is a clear and immediate danger to the train and its passengers, he shall continue to the next station before taking any action;
 - (ii) the Train Operator shall inform on radio the Traffic Controller in the OCC and Station Controller of the station at which the train stops about operation of the passenger emergency alarm;
 - (iii) in case of no passenger response for operated emergency alarm, the Train Operator can reset passenger emergency alarm with the approval of the Traffic Controller.
- (13) In case of poor visibility, rendering the sighting of signals difficult, the Train Operator shall advise the Traffic Controller and proceed in accordance with special instructions.
- (14) Where the train is driven manually in other than Coded Manual mode, the Train Operator shall ensure that the doors on the platform side only are opened for boarding and/or alighting the passengers.

33. Locomotives, work trains and maintenance vehicles —

- (1) Locomotives, work trains, and self-propelled maintenance vehicles equipped with ATP equipment shall be worked as per rules of operation for passenger trains.

- (2) The operator of a self-propelled maintenance vehicle which is permitted to operate on running lines shall hold a certificate of competency as a Train Operator or to be accompanied by a person holding certificate of competency and in the latter case, the person holding the certificate of competency is responsible for the observance of these rules in respect of the operation of the vehicle.
- (3) Before departing from the depot, or from a work site where the train has been uncoupled, the Train Operator of the leading locomotive shall—
 - (a) ensure that the train is fully coupled;
 - (b) carry out a continuity test of the pneumatic brake; and
 - (c) verify that all handbrakes have been released.
- (4) Subject to sub-rule (2), the Train Operator of a locomotive, work train or self-propelled maintenance vehicles, shall possess at all times, when on duty—
 - (a) a tri-colour torch capable of showing a red, green and white aspect; and
 - (b) any special notice relating to the working of work trains.
- (5) Any unpowered vehicles stationed on a siding or on the running line shall be secured by the application of sufficient number of handbrakes unless coupled to a locomotive.
- (6) Any such vehicle or group of vehicles shall have a lamp attached to an outermost vehicle displaying a red aspect in the direction of approaching trains and on a running line such lamps shall be placed at both the ends of the vehicle or group of vehicles.
- (7) Any self-propelled maintenance vehicle which is not fitted with ATP equipment shall be taken on running line only if, —
 - (a) during revenue service period, it is coupled to a locomotive or other vehicle which is so equipped and driven at such speed as specified in special instructions; or
 - (b) during revenue service period, it is working within the limits of engineer's possession; or
 - (c) during non-revenue service period, it is running on line of sight method at a speed in accordance with special instructions.
- (8) Shunting of vehicles to make or break work train shall only take place as per special instructions.
- (9) The maximum speed of vehicles, not equipped with ATP control, shall be limited in accordance with approved special instructions.
- (10) Fly shunting of any vehicle is expressly prohibited at any time.

Note: In this rule, a “fly shunt” is made when two vehicles are sent forward unattached either together or one immediately after the other and placed on different lines necessitating the points being reversed after the passage of the leading vehicle.

CHAPTER V

PLATFORM DOORS

34. **Kinds of platform doors** — Platform doors where provided, shall be of the following types, namely :—
- (a) Platform Screen Doors—these doors are powered doors located along the platform at the platform edge throughout the passenger area with door locations corresponding to the train car passenger door locations;
 - (b) Platform Edge Doors—these are powered doors located along the platform at the platform edge normally used for baggage;
 - (c) Manual Secondary Doors—these are manual doors located at one end of platform or both ends of platform to provide access between platform and the trackside;

- (d) Emergency Escape Doors—these doors are generally located around Platform Screen Doors of leading and trailing passenger cars and are meant for use in emergency situations;
- (e) Fixed Panel—platform length sections, not provided with any of the doors mentioned above viz Platform Screen Doors or Platform Edge Doors or Emergency Escape Doors or Manual Secondary Doors, are provided with fixed screens called Fixed Panel.
- (f) All the above doors and panels may be full height or half height as required.

35. Normal working of doors —

- (1) Platform Screen Door —
 - (i) Opening or closing of the Platform Screen Doors shall be after receipt of the DOORS OPEN or DOORS CLOSE command signals from the signalling link, or any other suitable arrangement;
 - (ii) signalling link or any other suitable arrangement enables automatic operation of the Platform Screen Doors only when the train stops within $\pm 500\text{mm}$ or the defined limit, as the case may be, of its normal stopping position. In the event of a train stopping outside this limit, the train may be driven in ATP, ROS, RM or cut-out mode under Train Operator's control, if necessary, to reposition it, and allow normal operation of the Platform Screen Doors when correctly positioned in accordance with sub-rules (8), (9), (10) and (11) of rule 32.
- (2) Platform Edge Door — Opening or closing of the Platform Edge Door where provided shall be by an authorised person, after receipt of indication of the DOORS OPEN or DOORS CLOSE command signals.
- (3) Manual Secondary Door — The Manual Secondary Door may be opened from the platform side by using a special maintenance staff key.

36. Abnormal working and emergency usages of doors —

- (1) Platform Screen Door —
 - (i) In case any Platform Screen Door does not open automatically due to any failure, the train passenger can go to the platform after opening the Platform Screen Door by using the manual release handle located on the track side of the door;
 - (ii) in case the Platform Screen Door does not open or close automatically after train doors opening or closing, then, the Train Operator or an authorised person may open or close all Platform Screen Doors using a local control panel located at the platform with the help of staff special key;
 - (iii) maintenance staff can open or close each individual Platform Screen Door from platform side using staff special key, if needed;
 - (iv) any defective Platform Screen Door may be isolated by using staff special key, if needed;
 - (v) in case it is not possible to close any Platform Screen Door, the speed of all trains entering the station, leaving the station or passing through the station shall be restricted to a maximum speed of twenty kilometres per hour and in addition, an audible warning shall be sounded by all trains while entering the station and further necessary precautions may be taken as per special instructions.
- (2) Platform Edge Door —
 - (i) In case Platform Edge Door local panel does not receive indication of the doors open or doors close command signals from the signalling link or from any other suitable arrangement, then the Train Operator or authorised person can provide this indication to local panel by using Platform Screen Door local panel with the help of staff special key and after receiving open or close indication, the authorised person can open or close the Platform Edge Doors;

- (ii) authorised person can open or close each individual Platform Edge Door using staff special key, if needed;
- (iii) any defective Platform Edge Door may be isolated by using staff special key, if needed;
- (iv) in case it is not possible to close any Platform Edge Door, the speed of all trains entering the station, leaving the station or passing through the station shall be restricted to a maximum speed of twenty kilometres per hour and in addition, an audible warning shall be sounded by all trains while entering the station.
- (3) Manual Secondary Door —
 - (i) In case of emergency evacuation from trackside, the Manual Secondary Door may be opened from the trackside by using a push bar;
 - (ii) the door shall be designed to swing open and be held at an open position of 90° and the door shall automatically revert to closed and locked position safely if it is left open at less than 90° without need for staff intervention.
- (4) Emergency Escape Door —
 - (i) In case train does not stop at the correct position and the opened train doors are not in front of Platform Screen Doors or under other specified conditions provided in special instructions, the passengers from the train can detrain to the platform after opening the emergency escape door by pressing the emergency push bar located on the track side of the emergency escape door;
 - (ii) the door shall be designed to swing open and be held at an open position of 90° and the door shall automatically revert to closed and locked position safely if it is left open at less than 90° without need for staff intervention.
- (5) Any door which has been manually released from track or platform side shall then be subjected to a gentle reclosing force, against which it can be pushed or held open, which shall ensure that the door returns safely to the closed and locked position afterwards, without need for staff intervention.

37. Indications —

- (1) Door open indicator —
 - (a) when Platform Screen Door or Platform Edge Door is under opening and closing process, the door opening indicator flashes and chime is activated;
 - (b) when Platform Screen Door or Platform Edge Door is under fully opened position, the door opening indicator is illuminated and chime is deactivated;
 - (c) when Platform Screen Door or Platform Edge Door is under closed and locked position, the door opening indicator is extinguished and chime is deactivated.
- (2) Indicators near the Manual Secondary Door or Emergency Escape Door— One each light emitting diode or other suitable indicator is provided near each Manual Secondary Door or Emergency Escape Door leaf to give closed and locked indications, respectively.
- (3) Obstruction detection —
 - (a) Sensors are provided to detect any person trapped between train door and Platform Screen Door at station with curved platform. When all Platform Screen Doors and Platform Edge Doors are closed by the Train Operator, the sensors shall check and give following indications on Platform Screen Door local panel at both headwall and tailwall:
 - (i) nobody trapped – Green; and
 - (ii) somebody trapped – Red flashing, unless sensors detect all doors clear, Platform Screen door system shall not give clearance for train departure;

- (b) in case of detection of an obstruction in between two leaves of a Platform Screen Door resulting in non closure of such door, the door makes further attempts to close automatically, if the obstacle is removed in these attempts Platform Screen door closes again, if obstacle is still present, then the Platform Screen door stops in unlocked position and is free to be pushed back by hand and once the obstruction disappears, the Train Operator can open and close the doors again and if the problem persists, then, station staff shall attend the affected door.

CHAPTER VI

CONTROL AND WORKING OF STATIONS

38. Class of stations —

The stations in metro railway are classified as under-

- (a) terminal stations; or
- (b) inter-locked stations having points and crossings, fixed signals and siding, or
- (c) other intermediate stations not having points and crossings (with or without fixed signals); or
- (d) any other class as specified in the special instructions

39. Responsibilities of Station Controller —

Every Station Controller shall —

- (i) open the station at least ten minutes before the scheduled time of the first train;
- (ii) carry out inspection of the station premises at the start of his duty period recording any defects or irregularities and reporting the same to authorities concerned including the OCC for prompt rectification;
- (iii) be responsible for the supervision of the passenger flow, and the provisions of adequate barriers and escalator services, ensuring that all staff renders prompt assistance to passengers;
- (iv) be responsible for reporting any defect and failure of any equipment on the station to the appropriate maintenance department;
- (v) be responsible for handling any emergent situations and public announcements;
- (vi) be responsible for the training of the station staff in local rules and conditions, for monitoring of their performances, discipline and administration;
- (vii) observe the departure of last train and at interchange station, shall supervise the interchange of passengers between the last advertised connecting train and shall inform the Traffic Controller when all interchange has been completed;
- (viii) inspect the station after the departure of last train to ensure that no unauthorised persons and unattended object remain on the premises and then lock all the entrances, as required;
- (ix) be responsible for keeping a log book which details occurrence on the station which shall include among other things, timings and reports of inspections, timing and location of maintenance activities, complaints or request from passengers, instruction from the Traffic Controller, period of local control of signalling and unusual incidents;
- (x) be responsible for control and operation of control panel or work station with the approval of the Traffic Controller;
- (xi) control the working of Platform Screen Door system as per special instructions;
- (xii) be responsible for management of TSRs and caution orders.
- (xiii) be responsible to other responsibilities as specified in these rules.

40. Responsibilities of platform supervisor —

The platform supervisor where provided, shall —

- (i) monitor the boarding and alighting of passengers and be alert to observe any accident and report to the Station Controller;
- (ii) when a dangerous situation arises, operate the emergency stop plunger to stop any train on or approaching the platform and report his action to the Station Controller;
- (iii) handle any malfunctioning of the Platform Screen Doors, where provided;
- (iv) manage customer interface;
- (v) monitor platform attendance;
- (vi) manage trolley circulation, if any; and
- (vii) be responsible for cleanliness of the platform.

41. Responsibilities of baggage handling supervisor — Wherever the baggage handling system is provided, the baggage handling supervisor shall—

- (i) monitor the entire baggage handling system to ensure smooth movement of passenger baggage; and
- (ii) take appropriate action as provided in special instructions in case of any abnormality in the baggage handling system.

42. Responsibilities of booking office staff —

(1) The booking office staff shall —

- (i) be responsible for the sale of tickets at ticket window and self-service machines where provided;
- (ii) sell tickets for the prices in the current fare table and render exact change as may be required;
- (iii) account for all tickets sold and all cash taken in accordance with instructions issued from time to time;
- (iv) keep such amount of cash at hispoints of sales, as is necessary for giving change and the surplus shall be kept in a locked safe or other secure storage;
- (v) be responsible for ensuring that the self-service ticket vending machines if provided, are adequately stocked with tickets and that cash is regularly removed from the machines to secure storage;
- (vi) be responsible for reporting malfunctioning or irregularities in the operation of ticket issuing equipments to the maintenance department; and
- (vii) assist the passengers during crowd control and emergent evacuation procedure.

(2) The senior booking office staff or assistant Station Controller, wherever provided, shall —

- (a) be responsible for the accurate accounting of the tickets sold and cash received; and
- (b) assist or depute for the Station Controller, when circumstances demand.

43. Security of stations—

- (a) Stations shall be open for access to the public at least ten minutes before the advertised time of departure for the first train and until all the passengers have left the station after the arrival of the last train and at all other times the station shall be secured, as laid down in the special instructions, against unauthorised entry;
- (b) at a location near station control room, a key to a designated emergency exit shall be provided in a glass box, which is for use of maintenance staff in the event of emergency during non-traffic hours. The emergency key box shall be inspected

each morning by Station Controller and use of the key, when made, shall be reported to the security controller who shall arrange for the box to be secured again;

The opening of emergency exit doors shall be monitored by suitable mechanism to keep track of its usage, and ensure their closure after use.

- (c) the emergency exits where ever provided, may also be used for the passenger's evacuation in emergency, if required.

2.

- (a) All the equipment rooms shall be kept locked at all times when access is not required and when these rooms are accessed by authorised person, such person shall be responsible for ensuring that no unauthorised person is permitted access.
- (b) All areas not required for the passage of passengers at the station shall be secured against unauthorised access;
- (c) Booking offices and other places where items of value, such as tickets and cash are kept, shall be locked at all times and within such areas tickets and other items of value shall be kept in safe storage.

44. Station working orders —

- (1) In addition to these rules and special instructions, each station shall be provided with the station working orders applicable to the station, giving details of-
 - (a) the location of equipment and guidelines for their use;
 - (b) the emergency evacuation routes at station with adjoining block section;
 - (c) the designated entrance for attendance by police, fire and ambulance vehicle;
 - (d) the designated entry for fire services; and
 - (e) list of medical facilities locally available.
- (2) Copies of these station working orders shall be readily available with the Station Controller on duty.
- (3) A copy of these working orders shall be kept in a special marked binder in a conspicuous place in the station control room.

45. Prevention of overcrowding —

- (1) In case a service delays or other incident causes a build-up of passengers on a platform, the Station Controller shall decide when that build up is likely to be unmanageable.
- (2) When any situation referred to in sub-rule (1) arises, the Station Controller shall reduce the flow of passenger to the platform by —
 - (a) making a warning announcement and stopping some or all of the inwards escalators;
 - (b) switching out some or all of the inward automatic fare collection barriers; and
 - (c) instructing the ticket sales staff to stop selling tickets.
- (3) If, despite all measure referred to in sub-rule (2) overcrowding develops in the concourse, passengers shall be advised to leave the station and the station entrances to be closed.
- (4) When train services are restored, the restrictive measures may be progressively or completely removed depending upon the frequency of train services available.

46. Emergency Evacuation —

- (1) The Station Controller shall control the evacuation from the station control room or from any other location as per special instructions, in the event of evacuation of the station

becoming necessary, as a result of cessation of train services, risk of fire or any other emergency –

- (a) all automatic fare collection barriers shall be set to open freely in the exit direction and the station staff deployed to assist in passenger evacuation;
 - (b) information and instruction shall be passed to the public by public address system and visual displays, where available;
 - (c) all inwards escalators may be stopped and used in the outwards direction, in accordance with special instructions;
 - (d) all ticket sales shall be suspended and the staff shall be used to assist in passenger evacuation; and
 - (e) all station exits shall be opened.
- (2) The station staff shall verify each area of the station that has been evacuated and thereafter when areas are verified as clear of passengers, the staff shall leave the station and secure it unless otherwise instructed by the Traffic Controller.
 - (3) In case fire or smoke is present, passengers shall, as far as practicable, be instructed to use exit routes that avoid contaminated areas.
 - (4) In case the fire is in the station, the Traffic Controller shall be informed so that he can instruct Train Operators not to stop the train at the station.
 - (5) In case the fire is at the concourse level, the train may be stopped to allow passengers to board only, as means of evacuating the passenger's more quickly and Train Operators shall make announcements in their trains to inform passengers not to alight from the train at the affected station.
 - (6) In case the fire is so extensive that the trains should not approach the station, the Station Controller shall inform the Traffic Controller to instruct the trains coming towards the affected station to stop at the previous station for evacuating the passengers there and this exercise shall prevail till normalcy is restored in the affected station.

47. Supervision of train movements —

- (1) When the station control is being exercised from the OCC, the Station Controller shall observe the passage of trains on the monitor and be alert to take action if the train service is in any way disrupted.
- (2) Any failure of any indication on the panel or work station or any other equipment shall be reported immediately to the Traffic Controller.
- (3) Control of the panel and workstation may only be taken with the permission of the Traffic Controller and the operation of individual routes and points shall be carried out as per the Traffic Controller's instructions

CHAPTER VII

ACCIDENT AND UNUSUAL OCCURRENCES

48. Report of accident and unusual occurrences —

- (1) Any accident or incident shall be reported by the authorised person concerned or any other person who notices it with utmost expediency to the Traffic Controller or the nearest Station Controller as soon as practicable.
- (2) On receipt of a report under sub-rule (1), the Traffic Controller shall inform the Station Controller and vice versa.

49. Duties of station staff or OCC —

- (1) On receipt of a report of an incident or accident or emergency, as the case may be, under rule 48, the Traffic Controller shall first ascertain the extent of injury to passengers and others and take prompt action to prevent further injuries and shall also assess the potential

effect on the train services and take all reasonable measures to maintain the train services, prevent delay or damage to property and equipment.

- (2) In an emergency, the Traffic Controller shall report it to the chief controller, who shall arrange for the assistance of the metro railway emergency response staff and where necessary, arrange the assistance of the Police, fire and ambulance services.
- (3) The Traffic Controller shall keep a log of all reports and requests received, action taken and other relevant information obtained or distributed.
- (4) The Station Controller, in the event of an accident at his station, shall take measures to prevent the situation becoming worse, render first aid if possible, arrange for the injured to be hospitalised and inform the Traffic Controller for outside help, if required.
- (5) In an emergency, the Station Controller shall evacuate the area concerned and take measures to prevent access to the area other than by the emergency services, and in extreme cases, the station may be closed and the Traffic Controller requested to arrange for trains to pass the station without stopping.
- (6) A full record of events and actions shall be entered in the station log, including the video recording as necessary, to preserve the clues.
- (7) All staff of the metro railway shall deal with accidents and emergencies expeditiously with the following priorities, namely:—
 - (a) save life, prevent further injury and alleviate suffering;
 - (b) protect the metro railway property and equipment;
 - (c) take steps for preservation of clues;
 - (d) inform the public of the effect on train services and the availability of alternative transport facilities;
 - (e) restore the safe operation of the train services as quickly as practicable; and
 - (f) restore normal services

50. Train stopped between stations —

- (i) In case the Train Operator cannot isolate a defect on his train and is unable to move it under its own power, he shall secure the train and request the Traffic Controller for assistance;
- (ii) the Traffic Controller shall, if possible instruct the Train Operator of the following train to drive as close to the stalled train as possible under Coded Manual mode and at the limit of authority under cab signalling;
- (iii) the Traffic Controller shall then instruct the Train Operator of the assisting train to change to RM mode or Cut out mode and to proceed at reduced speed and stop about ten meters short of the stalled train;
- (iv) in case, it is more convenient to provide assisting train from the leading end direction, the Traffic Controller shall instruct the Train Operator of the assisting train to detrain the passengers at the station, to change the cab and proceed in the direction of the stalled train under Coded Manual mode as far as limit of authority under cab signalling and thereafter change to RM mode and stop short of ten meters of the stalled train;
- (v) the Traffic Controller shall instruct the Train Operator of defective train to secure his train and instruct the Train Operator of the assisting train to couple the defective train by mechanical means only and to isolate all electrical connections to the defective train;
- (vi) once the trains are confirmed coupled as per special instructions, the Traffic Controller shall instruct the Train Operator of the defective train to release the brakes of his train;

- (vii) the Traffic Controller shall then authorise the Train Operator of the assisting train, if in front, to once again change the cab and drive forward in Cut-out mode at slow speed while exchanging communication with the front cab of defective train until the assisting train has completely reached the platform of the next station;
 - (viii) the combined consist shall then be moved forward until the defective train has completely reached the platform and there after the passengers of the defective train shall be detrained at the station and the combined consist shall be moved to the depot;
 - (ix) the Train Operator of the assisting train, if in the rear, shall drive the combined consist in Cut-out mode at a speed not exceeding ten kilometre per hour, while exchanging communication with the Train Operator of the defective train in the lead cab until the defective train is at the platform of the next station;
 - (x) all passengers shall be detrained from the defective train and the combined consist shall then be moved further until the assisting train is completely on the platform;
 - (xi) all passengers of assisting train shall then be detrained at the station and the combined consist worked to the depot or any suitable location in Cut-out mode at a speed not exceeding ten kilometre per hour, with leading and intermediate Train Operators exchanging communication on cab to cab telephone or train radio, particularly if being assisted from rear.
- (2)
- (i) In case traction power is lost, all trains shall coast as far as the momentum of the train and the signalling system permit, after ascertaining from Traffic Controller that there is no defect in the overhead contact wire or third rail system, as the case may be. The objective is to get every train to a platform or as close to the platform as possible where passengers can be detrained if the incident is likely to be prolonged;
 - (ii) in case the traction power has not been restored within fifteen minutes, passengers shall be detrained from all trains at stations and the process of detraining passengers from any train stopped between stations shall be initiated after switching off third rail power supply or overhead power supply in case of side evacuation, as the case may be, and the Train Operator shall also take suitable measures of securing his train;
 - (iii) the Traffic Controller shall advise the Station Controller of the adjoining station to which evacuation is planned, who shall in turn make suitable arrangement for assisting the evacuation process;
 - (iv) after evacuation of passengers, the Traffic Controller shall take appropriate action of working the stranded trains to nearest station siding or depot using other suitable shunting vehicles.
- (3)
- (i) In case a train cannot be moved as a result of derailment or a mechanical failure, leading to possible infringement, the Train Operator of the stalled train shall switch on the flasher light, secure the train, and immediately inform the Traffic Controller apprising him of any infringement on other track (if any) and possibility of injuries to passengers among other things and also inform him that he is unable to move his train;
 - (ii) the Traffic Controller in consultation with the Train Operator shall decide the most appropriate method of evacuation, assistance needed and working of failed train car taking into account, proximity of stations, availability of trains and other local conditions and advise the Train Operator, the direction from which assistance is to be provided and inform the Station Controller at the station to which passengers shall be evacuated;
 - (iii) the following methods of evacuation, as the case may be, shall be followed, namely: —
 - (a) evacuation to a train on the same track;

- (b) evacuation to a train on an adjacent track (except in twin tunnel sections);
 - (c) evacuation to train on adjacent track (in twin tunnels);
 - (d) evacuation on foot to the nearest station;
 - (e) evacuation by climbing down the viaduct or climbing up the evacuation shaft in tunnels;
- (4) Evacuation to a train on the same track:
- (a)
 - (i) Wherever considered necessary, the traffic on the adjacent track may be stopped, as specified under special instructions;
 - (ii) In case being assisted by an assisting train from the rear, passengers shall, if possible, be first detrained from the assisting train at station and the Traffic Controller shall instruct the Train Operator to drive in Coded Manual mode until the limit of authority under cab signalling;
 - (iii) the Traffic Controller shall, thereafter, instruct the Train Operator to change to RM mode or Cut out mode and drive at not more than twenty-five kilometre per hour to stop about ten meters from the stalled train;
 - (iv) in case assistance from the rear is not possible, assistance may be provided from the front end direction, following similar procedure once the Train Operator has changed the cab for driving in the other direction as specified in the special instructions;
 - (b) the Train Operator of the assisting train shall report to the OCC when he has reached this location and the Traffic Controller shall then instruct him to move his train slowly and stop it ten meters short of the stalled train;
 - (c) the Train Operator of the assisting train shall secure his train, and open the appropriate side or end door and ramps (if any), and the Train Operator of the stalled train shall open the corresponding door on his train and the two Train Operators shall assist passenger to shift from the stalled train to the assisting train;
 - (d) when all passengers have been transferred, the doors and ramps (if any), of both the trains shall be closed and secured;
 - (e) the Train Operator of the stalled train shall remain with his train and the Train Operator of the assisting train shall move to the other cab and report to the Traffic Controller that all passengers have been transferred and that his train is ready to move;
 - (f) only after receiving the instructions from the Traffic Controller, the Train Operator of the assisting train shall select RM mode or Cut out mode and drive the train to the station from which he came where passengers can be detrained.
- (5) Evacuation to a train on an adjacent track (except in twin tunnel sections):
- (a) In case assistance cannot easily be given by a train on the same track, a train on the adjacent track may be used;
 - (b) on receipt of a request for assistance, the Traffic Controller shall inform the Station Controllers at the station on the either side of the location of the incident;
 - (c) the passengers may be detrained from the assisting train at a station;
 - (d) the Traffic Controller shall instruct the Train Operator of the assisting train to select coded manual mode and proceed towards the stalled train as near as permissible and then, select RM mode or Cut out mode in consultation with the Traffic Controller and drive his train to a point near the stalled train;

- (e) the Train Operator of the assisting train shall secure his train and open the appropriate door of his train and report completion to the Traffic Controller;
 - (f) the Traffic Controller shall then instruct the Train Operator of the stalled train to open appropriate door and ramp (if any) of his train;
 - (g) the two Train Operators shall then supervise the transfer of passengers via the ramps or step ladders and the track from the stalled train to the assisting train, taking particular care to inform passengers of the dangers of tripping on rails and other equipment;
 - (h) once all passengers have been transferred, the doors and ramps (if any), shall be secured and completion reported to the Traffic Controller by the Train Operator of the assisting train;
 - (i) the Train Operator of the stalled train shall remain with his train;
 - (j) the Traffic Controller shall then authorise the Train Operator of the assisting train to proceed to the next station in the normal direction of traffic, where normal service may be resumed.
- (6) Evacuation to train on an adjacent track (in twin tunnels):
- (a) In twin tunnel sections, if assistance cannot easily be given by a train on the same track, a train on the adjacent track may be used;
 - (b) on receipt of request for assistance, the Traffic Controller shall inform the Station Controllers at the station on the either side of the location of the incident;
 - (c) passengers may be detrained from the assisting train at a station;
 - (d) the Traffic Controller shall instruct the Train Operator of the assisting train to select Coded Manual mode and proceed toward the direction of stalled train as near as permissible to an appropriate cross passage as advised by the Traffic Controller and then change to RM mode or Cut out mode in consultation with the Traffic Controller to drive his train to a point, near a cross passage as advised by the Traffic Controller;
 - (e) the Train Operator of the assisting train shall secure his train and open the appropriate door and ramp (if any), of his train and report completion to the Traffic Controller;
 - (f) the Traffic Controller shall then instruct the Train Operator of the stalled train to open its appropriate door and ramp (if any), as applicable, which is nearest to this cross passage;
 - (g) the two Train Operators shall then supervise the transfer of passengers via the cross passage and the track from the stalled train to the assisting train taking particular care to inform passengers of the dangers of tripping on rails and other equipment, and the hazards of walking through the cross passage, and the two Train Operators shall ensure that all passengers have been safely transferred and accounted for;
 - (h) once all passengers have been transferred, the appropriate doors and ramps (if any), of assisting train shall be closed, secured and completion reported to the Traffic Controller by the Train Operator of the assisting train;
 - (i) the Train Operator of the stalled train shall close the doors and ramps (if any), and return to his train cab;
 - (j) the Traffic Controller shall then authorise the assisting train to select appropriate mode and proceed to the next station where normal service may be resumed.
- (7) Evacuation on foot to the nearest station:
- (a) In case of evacuation on foot, the Traffic Controller shall decide as to which station passengers are to be evacuated which shall normally be the nearest station subject

- to other factors, such as the location of the trains, ventilation consideration in tunnel sections and any damage to track, train or structures which may make it desirable to use an alternative station;
- (b) the Traffic Controller shall inform the Station Controller at the station designated to receive the passengers and the Station Controller shall clear the platform concerned of waiting passengers and, if necessary, stop incoming passengers, and if a tunnel section is involved, he shall ensure that the tunnel lighting is switched on, open all emergency doors, Manual Secondary Doors and Platform Screen Doors, where provided, and he shall position himself and his security staff on the platform to receive the arriving passengers;
 - (c) the Station Controller shall prepare to render assistance or provide first-aid to any passenger who may have had difficulty or met an accident during the evacuation;
 - (d) on open sections and in double track tunnels, the Traffic Controller may arrange for traffic to be suspended on the adjacent track during the period of the evacuation and in case of third rail traction system, instruct the traction power controller to switch off the third rail power supply;
 - (e) the Traffic Controller shall verify with the Train Operator that the train has been secured and then instruct him to deploy the appropriate door and ramp (if any), of the train, nearest to the designated station;
 - (f) passengers shall be informed of the procedures to be followed and given explicit warning on tripping hazards, where to walk, or hazards of walking on the raised walkway, not to raise any object above head level and what to expect at the station;
 - (g) passengers shall be detrained on the track or on the raised walkway, as applicable by the Train Operator and directed to the station; and the Train Operator shall count passengers as they leave the train;
 - (h) the Station Controller shall count the passengers as they arrive at the platform;
 - (i) the Train Operator shall ensure that the last passenger has left the stalled train and also check that all passengers have left the track or raised walkway, as the case may be;
 - (j) the Train Operator and the Station Controller shall check their respective count of passengers' numbers and satisfy themselves that all passengers have reached the platform and thereafter the Train Operator shall return to his train and secure the end door and ramp (if any), or the side doors, as the case may be;
 - (k) the Station Controller shall record in the station log, the details of the incident; in particular, the number of passengers detrained, and then report the statistics to the Traffic Controller.
- (8) Evacuation on foot by climbing down on via-duct or climbing up the evacuation shaft in tunnels:
- (a) In case the train is near a suitable location designated for evacuation of passengers by climbing down the via-duct or climbing up the evacuation shaft in tunnels, the Train Operator of stalled train shall consult the Traffic Controller and on instructions from Traffic Controller, secure his train;
 - (b) the Traffic Controller shall intimate the Station Controllers of adjoining stations for assisting in the evacuation, who shall depute competent metro staff and security personnel at the safe evacuation point to assist the evacuation;
 - (c) the assisting metro staff shall render assistance or provide first-aid to any passenger who may have had difficulty or accident during the evacuation;
 - (d) the Traffic Controller may arrange for traffic to be suspended on the adjacent track for the duration of the evacuation;

- (e) on the advice of the Traffic Controller, the Train Operator of stalled train shall open the appropriate door or ramp (if any), of the stalled train;
 - (f) passengers shall be informed of the procedures to be followed and given explicit warning on tripping hazards, where to walk and not to raise any object above head level and what to expect at the evacuation point;
 - (g) passengers shall be detrained on the track or walkway by the Train Operator and directed to the evacuation staircase or rescue vehicle and the Train Operator shall count passengers as they leave the stalled train;
 - (h) the passengers shall again be counted as they arrive at the safe location after evacuation by the assisting metro staff at the safe evacuation point;
 - (i) the Train Operator shall ensure that all passengers have left the stalled train and check that all passengers have left the track and /or walkway;
 - (j) the Train Operator and the assisting metro staff at the safe evacuation point or vehicle shall check their respective count of passengers' numbers and satisfy themselves that all passengers have reached the safe evacuation point or vehicle and report the same to the Traffic Controller. The Train Operator shall, thereafter return to his train, secure the end door and seek further instructions from the Traffic Controller;
 - (k) the evacuated passengers shall be brought to the nearest station;
 - (l) the concerned Station Controller shall record in the station log the details of the incident, and in particular the number and other details of passengers detrained, and transfer of their luggage and repeat the complete incidence details to the Traffic Controller.
- (9) In case a train cannot be moved as a result of failure of traction system the Train Operator shall consult the Traffic Controller who in turn shall consult the traction power controller, and after ensuring that it is not possible to restore traction power, shall arrange to cut off traction power in the affected section and take any further measures for safety of passengers in accordance with sub-rules (5) to (8), wherever applicable, shall be taken to evacuate the passengers from the stalled train.
- (10) Notwithstanding, anything contained in sub-rules (1) to (9), in the case of third rail traction system, the third rail power supply shall be 'switched off' in all cases of evacuation in which passengers are required to come on the raised walkway or the track.

51. Train divided —

- (1) In case a train is stopped by application of an irrevocable emergency brake and cab signalling indications are normal, the Train Operator shall examine the TCMS panel, or TIMS panel to ascertain the cause, if indication of faults in multiple circuits affecting the whole train or rear cars of the train are present, the train shall not be moved until, it has been verified that the train is complete and coupled.
- (2) After the verification about complete arrival and the integrity of the train is completed under sub-rule (1), the Traffic Controller may authorise the Train Operator to make appropriate isolations and proceed. If the train is not divided, but is still unable to move on its own, the train shall be worked in accordance with sub-rule (1) of rule 50.
- (3) In case the train is found to have parted, the Train Operator shall first satisfy himself that no passenger has been injured or has fallen from the train, and then -
 - (i) the passengers shall be cleared of the open ends of the train and train re-coupled with the help of assisting staff as per special instructions, if possible in consultation with the Traffic Controller and rolling stock supervisor;
 - (ii) in case the Train Operator has successfully re-coupled the train, he shall return to the leading cab, report the circumstances to the Traffic Controller and seek

permission to proceed to the next station following the instructions given by the Traffic Controller.

- (4) The train shall be withdrawn from passenger service at the next station and worked to depot for investigation of the incident.
- (5) In case the train cannot be re-coupled, the Train Operator shall inform the Traffic Controller and further action shall be taken as per special instructions.

52. Unusual occurrences —

- (1) The authorised persons shall be conversant with the location and use of fire alarms and fire fighting equipment at their place of work.
- (2) The authorised persons observing the smoke or fire shall raise the alarm by means of the equipment provided or by informing the Station Controller and Traffic Controller, as may be most appropriate and expeditious.
- (3) In case smoke or fire is reported on a train between stations, the Train Operator shall inform the Traffic Controller, drive his train to the next station and detrain passengers. Traction power shall then be switched off, and in overhead traction territory, the pantographs of the affected train lowered, or in the third rail traction territory the current collection devices of the affected train retracted, before traction power is restored to other trains.
- (4) In case the fire on a train or on the track causes a train to stop between the stations, passengers shall be evacuated as per the provisions specified in sub-rules (3) to (8) of rule 50
- (5) In case the incident occurs in a tunnel, the Traffic Controller shall arrange with the auxiliary systems controller for the ventilation system to supply fresh air to the chosen route for evacuation before authorising detrainment of passengers.
- (6) In case the fire alarm on a station is actuated or a verbal report is received of smoke or fire in the station, the Station Controller shall inform the Traffic Controller and then verify himself by closed circuit television or actual inspection, whether or not the alarm is genuine.
- (7) In case smoke or fire is present, the Station Controller shall inform the Traffic Controller and arrange for passengers to be evacuated from the area concerned preventing further access and if necessary, the station may be completely evacuated and the Traffic Controller may be requested to arrange for trains not to stop.
- (8) The Traffic Controller shall inform the chief controller who shall arrange for the fire fighting services and if necessary, the ambulance services.
- (9) In case a Train Operator or Station Controller observes a fire in adjacent premises which would affect the property of the metro railway, he shall report it to the Traffic Controller and the Traffic Controller shall inform the chief controller and the security controller and maintain normal services unless or until a local inspection confirms that a potential danger exists.

53. Flooding —

- (1) The Train Operator or Station Controller or the other member of the staff, who observes water accumulating on the track shall report to the Traffic Controller giving as much detail as possible with respect to location, distance of track affected, and approximate level of water with respect to the rail.
- (2) The Traffic Controller shall inform all trains required to pass through the area and request reports of the water level and if the water level is below the level of rail fastenings, the Traffic Controller shall instruct the Train Operator to reduce the speed of their trains to twenty-five kilometre per hour when passing through the affected area.

- (3) In case the water level rises above rail fastenings passenger train service shall only be permitted under special instructions.

54. Other unsafe conditions —

- (1) All authorised persons, and in particular the Train Operators and Station Controllers shall keep a look out for unsafe conditions on or in the vicinity of the railway track which are as follows: -
 - (a) damaged or dislodged fixed equipment within the railway right of way;
 - (b) broken or buckled rails;
 - (c) displaced or damaged overhead traction power conductors or third rail power conductors, as the case may be;
 - (d) construction activities adjacent to the track including use of cranes which can swing within 6 metres of the track;
 - (e) road accidents which might cause or have caused damage to bridges and viaducts;
 - (f) road accidents which might cause or have caused vehicles or their loads to encroach on the metro railway right of way; and
 - (g) any other obstruction on the track.
- (2) In case the Train Operator observes any unsafe condition, he shall report to the Traffic Controller immediately so that action can be taken to minimise the effect and remove the cause.
- (3) In the event of an earthquake, the Traffic Controller shall instruct all trains to stop immediately and after the earthquake has subsided, the Traffic Controller may instruct each stranded Train Operator to proceed in RM mode or Cut out mode at walking speed after examining that the track is safe for train movement and free from obstruction to up the next station:

Provided that in such event, the normal operation of trains may be resumed, if all the track and structures are examined as per special instructions.

55. Accidents —

- (1) In case of accidents, arrangements for medical aid, evacuation of sick, injured passengers, access for ambulance, staff and vehicles shall be made and included as per provisions specified in special instructions.
- (2) In the event of serious accident, the chief controller may, in consultation with senior management declare an emergency as per the provisions specified in special instructions.
- (3) A senior member of the management shall be appointed as an emergency officer and shall set up an emergency control either at OCC or at the site depending on the nature of the occurrence.
- (4) The emergency officer shall be in overall charge of all the metro railway resources of staff and materials for the handling of the emergency and the coordination between the metro railway and external emergency agencies, such as fire, ambulance and police and utility services.

CHAPTER VIII

SYSTEMS OF WORKING

56. CATC System —

- (1) The CATC system of working shall be adopted on the metro railway, for movement of trains between stations and between depot and the main line.
- (2) The CATC system which includes CBTC works on the principle of target speed and target distance with Cab Signalling by means of continuous transmission between trackside and

train through suitable approved means, ensuring safe movement of all trains under all operating conditions by continuously generating a safe operating envelope defined by the limit of movement authority and the maximum safe speed.

- (3) The limit of movement authority shall be the farthest point to which the train may safely proceed taking into account margins for error in speed and distance measurement, calculating braking distances and the equipment reaction times.
- (4) The maximum safe speed shall be the speed at which the train is permitted to travel without intervention by the train control and signalling system and it shall be continuously calculated in such a manner that PSRs, the speed limits for the type of train and TSRs shall not be exceeded and the train shall always stop without passing the limit of movement authority.
- (5) The CATC system shall provide the following modes of train operation, namely: –
 - (a) UTO;
 - (b) DTO;
 - (c) AM, where provided;
 - (d) automatic reversal mode or automatic turn back mode, if provided;
 - (e) coded manual mode;
 - (f) RM Mode;
 - (g) Cut Out Mode; and
 - (h) ROS Mode, where provided

57. Automatic Mode —

- (1) In the AM, the train shall operate without intervention of the Train Operator except closing of train doors and starting from a station stop and it shall operate under the supervision and control of the ATP functions.
- (2) In AM, the train control and the signalling system shall—
 - (a) accelerate and decelerate the train by applying traction power, coasting and applying and releasing brakes;
 - (b) automatically control speed, acceleration, and stop the train at stations;
 - (c) provide all indications necessary to operate the train;
 - (d) determine continuously the maximum safe speed and limit of movement authority;
 - (e) prevent movement of the train in excess of the maximum safe speed and limit of movement authority ;
 - (f) open train doors on the correct side when the train is docked if permitted by the ATP door release and the Platform Screen Doors, where provided open automatically on the correct side;
 - (g) prevent the train from starting if train doors or the Platform Screen Doors, where provided are not detected closed;
 - (h) train re-starting from a signal stop shall be automatic; and
 - (i) train starting or re-starting from a station stop shall be initiated by the Train Operator.

58. Automatic Reversal Mode or automatic turn back mode; —

- (1) This mode, where provided, is used to reverse the running direction of a train automatically in areas of the section specifically specified in the special instructions when train is at standstill.
- (2) The transition from AM or coded manual mode to automatic reversal mode is initiated automatically upon receiving the request for reversal operation from ATS and has to be acknowledged by the Train Operator by pressing the automatic reversal button at standstill.

- (3) The on-board ATP unit of leading cab activates the unit at trailing end on arrival at a station, if a reversal operation is requested by the Train Operator.
- (4) The train borne ATP unit shall return from automatic reversal mode to coded manual mode once the reversal operation has been carried out successfully and the Train Operator has unlocked the new leading cab for further running.

59. Coded manual mode —

- (1) In coded manual mode the train shall be driven by the Train Operator, obeying cab signals.
- (2) The train control and signalling system shall ;-
 - (a) provide cab signals and all other indications necessary to operate the train including current speed;
 - (b) determine continuously the target speed and limit of movement authority;
 - (c) prevent train operation in excess of the target speed or limit of movement authority;
 - (d) provide audible and visual warning if the train speed exceeds the target speed or the maximum safe speed;
 - (e) enable train doors and platform screen doors wherever provided, when the train is docked, enabling only the doors on the platform side of the train; and
 - (f) prevent the train from starting, if train doors or the Platform Screen Doors where provided are not detected closed.

60. RM Mode —

- (1) The RM Mode is the default mode of operation and is automatically initiated, when the automatic train control train borne equipment is first powered and it remains in operation until sufficient conditions have been met to allow for a transfer to coded manual or AM.
- (2) The RM Mode shall be used in the absence of coded manual or AMs-
 - (a) to operate trains in depots;
 - (b) following an emergency brake application on main line and absence of cab signals;
 - (c) for entry to and upto exit from the depot; and
 - (d) on instructions from the Traffic Controller.
- (3) In RM Mode, the train speed shall be limited to a maximum of twenty-five kilometre per hour enforced by on board ATP equipment.
- (4) All Platform Screen Doors, where provided shall open automatically or have to be operated manually by the Train Operator or an authorised person from the local control panel provided near the train cab at the platform.
- (5) The train doors on the correct side shall also have to be operated manually by the Train Operator.

61. Cut-out-Mode —

- (1) The 'Cut Out' Mode, is intended for use in case of complete failure of train borne signalling and train control system and in such mode, the train speed shall be restricted to twenty-five kilometre per hour, unless special provision is made as under-
 - (i) Low Speed Cut-Out-Mode —this is the default mode of operation in automatic train control cut-out condition, in which the propulsion system would cut out traction of the train above twenty-five kilometre per hour speed and the Train Operator shall manually drive the train in accordance with the line-side signals or as per special instructions and shall monitor and limit the speed of the train if it exceeds twenty-five kilometre per hour speed, by service brake or emergency brake application.

- (ii) High Speed cut-out-mode —this mode where provided, shall be employed only under specific instructions from the OCC, in which the operation in automatic train control cut-out condition, the propulsion system would cut out traction of the train above forty kilometre per hour and the Train Operator shall manually drive the train in accordance with the line-side signals or as per special instructions and shall monitor and limit the speed of the train if it exceeds forty kilometre per hour, by service brake or emergency brake application.
 - (2) In cut-out-mode, the train shall be operated by the Train Operator in accordance with line side signals and on radio verbal instructions from the Traffic Controller.
 - (3) Running of trains on mainline in Cut Out Mode is permitted only under instructions of the Traffic Controller.
 - (4) All Platform Screen Doors, where provided, shall have to be operated manually by the Train Operator from the local panel provided near the train cab at the platform.
 - (5) The train doors on the correct side shall also have to be operated manually by the Train Operator.
62. **Run On Sight (ROS) Mode** — In ROS Mode, which is a sub-mode to the Coded Manual Mode and only operates in the absence of ATP signals from train control ground equipments, the train is driven manually on line of sight and the speed is limited by ATP system to a maximum of twenty-five kilometre per hour. When ATP signals from train control ground equipments are received, this mode automatically changes to Coded Manual Mode.

CHAPTER IX

SINGLE LINE WORKING

63. **Adoption of single line working in certain cases -Single line working shall be adopted in the following situations, namely: —**
- (1) normal operation on a single line or twin single lines shall be adopted when the signalling system permits operation in both directions, and such operation shall be carried out as per procedure specified under sub-rule (1) of rule 64.
 - (2) When there is an obstruction on a portion of line between any two stations on one of the running lines, the train services may be continued in the affected section on the adjacent line in both directions using single line working on the unaffected track and such single line working shall be operated in the manner as specified under sub-rule (2) of rule 64.
64. **Implementation—**
- (1) For the purpose of normal operations on single line or twin single lines under sub-rule (1) of rule 63, the following procedure shall be followed namely: -
 - (i) the train shall operate under the authority of cab signals under Coded Manual Mode or AM (if signalling system permits) on the single line or on the respective line of twin single lines;
 - (ii) in case of failure of cab signals under Coded Manual Mode or AM, the trains may be allowed to run under the RM Mode, ROS Mode or Cut Out Mode, under Special Instructions.
 - (2) For the purposes of sub-rule (2) of rule 63, where a portion of a line is obstructed on any of the running lines, the Traffic Controller may, after consultation with the chief controller decide to implement the single line working in the following manner, namely:-
 - (i) single line working on the unaffected track shall be achieved by fleeting of trains in groups in one direction at intervals decided by the Traffic Controller followed by a group of trains in the other direction;
 - (ii) the Traffic Controller shall before the train passes in the reverse direction, inform to the Station Controller at each of the stations in the single line section;

- (iii) the Station Controllers shall also closely monitor crowding on the platform in use and prepare to close entrances to the station if overcrowding becomes dangerous;
- (iv) the Station Controllers at these stations shall inform passengers by visual and audible announcement and take such other measures as may be necessary to direct passengers on the correct platform;
- (v) the Traffic Controller shall inform the Train Operators of all trains of the location and direction of the single line working;
- (vi) trains shall work under normal signalling including ATO in the normal direction of travel and in the case of reverse direction of travel, if signalling system permits, the train shall normally work under AM or Coded Manual Mode and other modes like RM Mode, ROS Mode or Cut Out Mode may also be used for train operation in the reverse direction of travel, under special instructions;
- (vii) when normal working is to be restored, the Traffic Controller shall inform the Station Controller at each station of the single line section of the last train to pass the section in the reverse direction after which normal station operation shall be restored.

65. **Signalling Failures** —In the event of failure of a track side signalling equipment which makes RM operation necessary on all trains, single line working under sub-rule (2) of rule 64 shall be suspended immediately and if necessary, train may still pass through the section in the normal direction of travel and after defect is repaired; single line working may be resumed.

CHAPTER X

PERMANENT WAY AND WORKS

66. **General** —

- (1) All running tracks shall be inspected as per the schedule specified in the special instructions.
- (2) All authorised persons whose duties require them to go on the tracks shall be properly trained.
- (3) All authorised persons who go on the tracks shall wear appropriate high visibility clothing, helmet and boots.

67. **Track work and track side work in non-traffic hours** —

- (1) No maintenance staff shall enter onto the track of any running line without the permission of the Traffic Controller.
- (2) Non traffic hours are defined as the hours between the passage of the last train, including any work train and a published time before start of traffic again in the morning and the normal time shall be published in the relevant handbooks which may be varied from time to time by the metro railway.
- (3) The maintenance staff requiring to carry out inspection or repair of equipment which does not affect the integrity of the track nor require the use of ladders or scaffolding and not involving danger to trains or traffic, may enter on the track under the following conditions, namely: -
 - (a) the Traffic Controller shall give permission, specifying the location and area for which permission is given and the time by which staff shall have to leave the track;
 - (b) the Traffic Controller shall log the time, location and the name of the person to whom permission under clause (a) has been given;
 - (c) on completion of the work, the person to whom permission has been given under clause (a) shall report to the Traffic Controller, identify himself and affirm that he and his equipment are clear of the track and that it is safe for service to resume;

- (d) in case the work cannot be completed within the allotted time, the person to whom permission has been given under clause (a) shall inform the Traffic Controller before the expiry of the time he has been allotted and request the Traffic Controller for an extension of time and the institution of an engineer's possession; and
 - (e) the Traffic Controller shall not permit the start of normal service until all permissions to work have been properly given up and rescinded.
- (4) All other works carried in non-traffic hours shall be protected by an engineer's possession.
- 68. Track work and track side work which extends into traffic hours —**
- (1) All works which are planned to extend beyond non traffic hours into the hours when train services normally operate shall be notified at least a fortnight in advance.
 - (2) All such works shall take place within an engineer's possession.
 - (3) Works which are planned to be carried out within non traffic hours without an engineer's possession but are delayed by unforeseen circumstances shall be protected by an engineer's possession.
 - (4) The train services on adjacent sections of the line shall be operated with appropriate caution order or other precautions as specified in special instructions.
- 69. Emergency track work or track side work in traffic hours —**
- (1) No routine maintenance shall be undertaken during the hours in which train services normally run except as provided for in sub-rule (2).
 - (2) In case emergency repair work is required to be carried out to prevent accidents or to maintain or restore train services, such emergency work shall be done under the engineer's possession which shall be granted by the Traffic Controller without delay taking the exigencies of trains services into account and making adjustments in the train schedules.
- 70. Engineer's possession —**
- (1) Engineer's possession on running lines is granted by the Traffic Controller who has final responsibility on whether or not the engineers may take possession.
 - (2) All works on tracks in depot shall be undertaken within an engineer's possession which shall be granted by the depot controller but in other matters the provisions hereinafter provided shall apply.
 - (3) An area under the engineer's possession is the sole responsibility of engineering official-in-charge and all issues of safe working within that area including the movement of trains is his responsibility.
 - (4) The person-in-charge of engineer's possession shall be trained in the duties and responsibilities of the role and certified by the authorised official or institution of metro railway.
 - (5) In case more than one maintenance unit is working within the same possession, one person shall be nominated by the Traffic Controller as the person responsible for the coordination of the work of all the units, as per special instructions.
 - (6) When possession is granted under sub-rules (1) and (2), the engineering official- in- charge shall protect the area of the possession from access by trains in one of the following ways, namely: -
 - (a) securing a vehicle at the limit of the possessing;
 - (b) securing points for a route which diverts trains away from the area of possession;
 - (c) using visual warning means for protection; and
 - (d) any other means as per special instructions.
 - (7) For running line signalled for operation in either direction, protection shall be implemented at all points of potential access and different methods may be used at each location in accordance with the sub-rule (6).

- (8) In case a work train or self-propelled maintenance vehicle is to be used within the possession area, it shall arrive at site before possession is taken and lamps displaying red aspects restricting movement towards the possession area shall be placed at the limits of the possession or on the secured vehicle where these are used to protect the possession area.
- (9) Where work on one track is likely to affect the passage of trains on an adjacent track, possession shall be taken of all tracks likely to be affected.
- (10) In case trains are required to pass on an adjacent track, the person-in-charges of the possession shall be responsible for ensuring that the track is safe to use before giving permission by hand signal.
- (11) Unless essential for the movement of trains, traction power shall be switched off from the area of the possession by the traction power controller and shall only be re-energised on receipt of clearance.
- (12) All switching ON or switching OFF operations of traction power required for maintenance work or engineer's possession or for any other purpose shall be approved by traction power controller.
- (13) The person-in-charge shall be responsible for confirming to the Traffic Controller on completion of the work that the track is safe for traction power to be switched on, all protection measures have been removed and the track is safe for trains to run.
- (14) All engineer's possession shall normally be authorised only under staff protection keys, where provided.

71. Works at stations —

- (1) No maintenance work affecting safety of train operation shall be carried out at any station without engineer's possession and until permission has been granted by the Station Controller.
- (2) All works in public areas shall be securely fenced to prevent access by the public.
- (3) The maintenance staff, as per special instructions, shall report to the Station Controller before starting work and again before leaving the station and the Station Controller shall record the time and location of their work so that they be warned of any emergency arising on the station.
- (4) Any maintenance work on a station which requires fire alarm or fire suppression equipment to be isolated shall be done only with the permission of the Station Controller.
- (5) The maintenance staff shall be responsible for their own protection and for raising the alarm in case of fire in the area which has been isolated and shall also be responsible for restoring normal function to the isolated equipment informing the Station Controller on completion of their work.
- (6) No maintenance work which requires the complete shutdown of the fire alarm or fire suppression systems of public areas shall take place during the hours in which the station is open to the public.
- (7) No maintenance work, which requires the use of ladders or scaffolding, shall take place within the fixed structure dimensions laid down in the schedule of dimensions from the platform edge during traffic hours.

CHAPTER XI

POWER SUPPLY AND TRACTION ARRANGEMENT

72. Switching on, and off, of traction and power supply distribution —

- (1) All traction and power distribution systems shall remain live at all times and shall only be switched off when necessary for maintenance of the equipment or protection of other maintenance activities.
- (2) Traction power supplies shall be switched on and off as per instructions of the Traffic Controller or depot controller (in his area of jurisdiction):

provided that in case of emergency after giving information to the Traffic Controller and the depot controller as the case may be, Traction Power Controller shall be authorised to switch off the power.

- (4) All traction and power distribution systems which have been switched off shall be adequately earthed before any maintenance or repair work is undertaken.
- (5) All maintenance work on high tension distribution network shall be undertaken with the permission of the traction power controller who shall ensure maintaining adequate power supplies for the operation of the metro railway network under the condition prevailing at that time.
- (6) Emergency trip system –
 - (a) The emergency trip switches shall be used for switching off the traction power supply in a specified section in case of any emergency, as per special instructions keeping in view that approaching train may apply emergency brakes and still enter the affected area.
 - (b) The emergency trip system switches are installed at each end of platform in stations, at station control rooms, traction sub-station and at cross passages in tunnelshaving third rail system. After activating an emergency trip system an individual signal is sent to SCADA system, and according to a tripping scheme and logic provided, the SCADA system shall switch off relevant circuit breakers and send signals to indicate that the track section is de-energized.
 - (c) In systems with overhead traction power, the requirement to disconnect traction power shall be permitted by approved alternative means.

73. Access —

- (1) All points of potential access by the authorised persons or public to high voltage equipment shall be kept locked and suitable warning notices displayed therein.
- (2) All switch gear and other high voltage equipment shall be fenced off and shall be accessible only to authorised staff and access to the live equipment shall only be permitted when the equipment has been switched off and earthed as laid down under special instructions.
- (3) No person shall work on high voltage equipment, or DC traction equipment or switch such equipment by local control unless he is properly trained and certified to perform such duties and possess a valid certificate of competency. He shall also obtain the permission of the traction power controller before undertaking such work and the procedure for obtaining, granting and cancelling of such permission shall be specified in special instructions.
- (4) No person shall work closer than two meters to a live conductor.

74. Arcing and fire —

- (1) In case any member of staff observes fire, smoke, arcing or fusing in the vicinity of the overhead line equipment or third rail equipment, as the case may be, he shall inform the Traffic Controller or Station Controller as quickly as possible and request power to be switched off from the section of the line.
- (2) The Traffic Controller shall instruct the traction power controller to switch off the power to that section of line before taking further action to investigate the occurrence.
- (3) The Traffic Controller shall also instruct the Station Controller at the nearest station to check if the fire or arcing has been extinguished.
- (4) On the basis of report of investigation and confirming rectification of the fault, traction power controller shall re-energise the section and may advise the Traffic Controller to resume train services.

75. Inspection and maintenance of electrical way and works —

The electrical way and work shall be inspected and maintained regularly in accordance with the provisions of the metro railway traction manual and special instructions.

76. Issue of caution order —

In case of breakdown of overhead equipment or third rail equipment, as the case may be, when it is necessary for a train to proceed cautiously, the traction power controller shall arrange for issue of caution order in accordance with the procedure in force.

77. Protection of the trains in case of traction system failure or breakdown —

Whenever a Train Operator finds that his train cannot proceed further on account of traction system failure or breakdown, the Train Operator shall follow instructions laid down in sub-rule (2) of rule 50.

78. Permit to work adjacent to or involving electrical equipment —

- (1) Works adjacent to electrical equipment or involving any parts thereof shall be carried out only by authorised person of the metro railway.
- (2) No work under sub-rule (1) shall be undertaken unless an authorised electrical person has been given a permit to work in accordance with the special instructions,
- (3) The authorised electrical person shall not issue such permit to work without the consent of the traction power controller who in turn, shall keep the Traffic Controller informed when train movements are affected. Whenever such work is likely to affect any other installations, the work shall be carried out as per special instructions.
- (4) Before commencing any work within two meters of overhead equipment or third rail or its associated cables and equipment, a permit to work shall be obtained from an authorised electrical person.
- (5) The maintenance and operation of electrical traction installations in emergency conditions shall be carried out after obtaining permit to work as per procedure specified in special instructions.

79. Work on service building and structures in the vicinity of live equipment —

The authorised person required to carry out work on service buildings and structures in the proximity of overhead or third rail equipment shall exercise special care to ensure that tools, measuring tapes, materials are not placed in a position from which they are likely to fall or make contact with electrical equipment.

80. Sectioning and siding switches —

- (1) Sectioning and siding switches installed in the traction system shall be operated only by such officials as are authorised to do so by the in-charge of the traction system.
- (2) No switch affecting the feed to main running line or loop line shall be closed or opened without permission of the traction power controller and the detailed procedures for opening and closing of sectioning and isolating switches shall be as per the metro railway traction manual.
- (3) All operations of section or isolating switches when completed, shall be reported to the traction power controller in all cases.

81. Warning to staff and public —

- (1) All electrical equipment shall be regarded as being live at all times and consequently dangerous to human life, save and except in cases where the electrical equipment has been specially made dead, isolated and earthed as per instructions contained in the metro railway traction manual.
- (2) Caution notices, in Schedule I, shall be prominently and permanently fixed near all vulnerable places to warn staff and public to exercise due caution.

- (3) No person shall climb on the top of rolling stock or go under the rolling stock in case of third rail DC traction, when these vehicles are located beneath the overhead equipment or adjacent to the third rail except when the overhead equipment or third rail is made dead and earthed in accordance with special instructions and the person required to climb on the roofs of passenger cars or to go under the same, shall obtain a permit to work as per rule 78.

82. Alterations to track —

Before any alteration to alignment or level of electrified tracks is commenced, due notice shall be given to those responsible for the overhead equipment or the third rail or signalling & train control system and all other concerned, as the case may be, so that the overhead equipment or third rail may be adjusted using specified gauge to conform to the new conditions of track.

83. Tunnel ventilation system and additional rules for tunnel ventilation & electrified sections —

- (a) For all underground sections, there shall be tunnel ventilation system equipped with suitable type and size of fans with the following functionalities to meet requirement of fresh air as well as handling of fire in the tunnel, namely:-
- (i) its fans may be operated locally or remotely;
 - (ii) provide fresh air;
 - (iii) extract smoke in one direction and supply air from other end of the tunnel section;
 - (iv) as far as possible, mode selection of extraction of smoke and/or pumping air from other end may be operated by single command from OCC or from the respective stations in case of failure of OCC;
- (b) special instructions for tunnel ventilation system shall be issued;
- (c) special instructions, if any, for working of trains on electrified sections shall be issued.

84. Rules applicable to overhead AC traction equipment only —

- (1) Tripping of circuit breakers of train in neutral sections —
- (i) Unless otherwise allowed by special instructions, the Train Operator shall coast through the neutral section, duly switching off power and necessary indications boards in Schedule II, wherever required, shall be provided to guide the Train Operator to switch off and switch on the power.
 - (ii) The Indication boards shall be provided at five hundred meters and two hundred and fifty meters in advance of the neutral section and additional boards shall be provided just short of and immediately after the neutral section to indicate to the Train Operator the points where he shall open and re-close the circuit breaker on the electrical multiple unit.
- (2) Tower wagon or catenary inspection car —
- (i) the movement or working of tower wagon or inspection car or any other vehicle with pantograph raised shall be as per the special instructions;
 - (ii) notower wagon or inspection car shall be driven except by an authorised person and no person shall be so authorised, unless he has knowledge of the section on which tower wagon or inspection car is to operate in addition to being conversant with the operation of tower wagon or inspection car.
- (3) Working of ladder trollies —
- (i) ladder trollies shall be considered as work trains and their movement on the main line track shall be governed by rule 33;
 - (ii) the ladder trollies shall work during Engineer's Possession only and their operation shall be supervised by a suitable traction distribution official as prescribed in special instructions.

85. Rules applicable to DC traction third rail equipment only –Switching ‘OFF’ and ‘ON’ of traction power-

- (1) Traction power shall ordinarily remain ‘ON’ and shall be switched ‘OFF’ during traffic hours for a specified section by the traction power controller or through the authorised person immediately after informing the Traffic Controller in emergencies, like: —

- (a) to stop serious arcing or fusing;
- (b) to stop train in unusual circumstances; and
- (c) when flood water overflows the track.

The traction power controller shall issue emergency power block as per prescribed procedure, to the authorised person. The Traffic Controller shall also inform all concerned that train movements on the affected section have been stopped.

- (2) When the cause of ‘Switching OFF’ is over, the authorised person who has taken emergency power block shall inform traction power controller to that effect and cancel the emergency power block.
- (3) Before switching ‘ON’ the current, the traction power controller shall obtain the permission of Traffic Controller who shall confirm from the engineering and signalling officials that the track is safe and free from any obstructions before permitting energisation of third rail. Thereafter, Traffic Controllers shall resume the traffic.

86. Procedure for preventing admission of electric rolling stock into or over sections of track with dead or earthed overhead lines or third rail —

- (1) In order to prevent electric rolling stock from being admitted into a track or a crossover for which overhead equipment or third rail equipment, as the case may be, is made dead or for which a permit to work is to be issued, suitable measures shall be enforced to block setting of such routes.
- (2) The levers or slide or push buttons or other means as provided for signals and points governing movement of electric rolling stock shall be suitably protected and if the points and signals are locally operated they shall be clamped and padlocked in their normal position and the keys shall be kept with the Station Controller or depot controller as the case may be.
- (3) These protective measures shall not be withdrawn until the Station Controller or the depot controller, as the case may be, receives a message from the Traffic Controller and acknowledges the same, and the Traffic Controller shall not issue a message unless he has received a message from the traction power controller cancelling the power block.

CHAPTER – XII

DRIVERLESS AND UNATTENDED TRAIN OPERATION

87. Unattended Train Management System (UTMS) —

- (1) The UTMS shall provide the following two modes of train operation, namely: —
- (a) “Driverless Train Operation (DTO)” which means a train operation where a Roving Attendant is on-board the train, but normally not in the driving cab, and his or her duties shall be specified as per special instructions;
 - (b) Unattended Train Operation (UTO) Mode where Roving Attendant or ETO or Train Operator may or may not be on-board as per special instructions.
- (2) The UTMS shall provide following system capabilities, namely: —
- (a) start, acceleration, deceleration and stoppage of train at stations, train doors and Platform Screen Doors where provided, shall be opened and closed, trains and their status from main line and depot shall be tracked and supervised including delay information;

- (b) wake-up and sleep process including safety tests of train at wake-up;
 - (c) automatic routing of trains to and from stabling as per time table;
 - (d) safe passenger transfer to prevent injuries to persons between cars or between platform and train;
 - (e) various measures to prevent collision with obstacles and persons on track by means such as physical barriers on guide way for elevated section and controlled access to tunnels;
 - (f) prevent access to track by unauthorised persons by use of Platform Screen Doors or other suitable means including physical barriers described earlier; Metros may also use operational practices as per special instructions;
 - (g) detection and management of emergency situations by performing train diagnostic, detection of fire and/or smoke, detection and handling of emergency situations (call or evacuation, supervision) by system and or staff in OCC including Roving Attendant and station staff or any other authorised person;
 - (h) alarms and messages for failure of equipment to OCC, remote commands to manage disruption to passenger services with support of OCC and station staff or any other authorised person;
- (3)
- (i) transfer indication of Passenger Emergency Alarm (PEA) operation and means of communication between OCC and passenger(s) on train;
 - (j) CCTV cameras inside the train and in front where provided, may be used by OCC for live video of interior or rear or front of a train.
 - (i) A Roving Attendant who is on-board a DTO or UTO train shall perform the following duties, namely:-
 - (a) be present in the train and remain watchful;
 - (b) be on general lookout from front window of the cab;
 - (c) report any unusual occurrence to the Traffic Controller;
 - (d) take over control of train and perform duties of Train Operator when required in accordance with rule 32; and
 - (e) do any other duty as assigned by general or special order of the metro railway administration or as instructed by the Traffic Controller.
 - (ii) In addition to the duties mentioned in these rules, a metro railway employee or an authorised person on noticing or becoming aware about changing rail conditions resulting to low adhesion shall report such matter to the OCC.
 - (iii) On receipt of report or becoming aware through alarm, where provided, necessary action shall be taken by the concerned person, as per special instructions, and check such areas which are observed to be affected by grease, oil, water or other substance, which could cause running rails to become slippery.
 - (iv) Traffic Controller may impose TSR in such areas as per special instructions.
- (4) The duties of Train Operator, OCC, Station Controller, metro railway employee or authorised persons and Depot Controller, shall be applicable in DTO or UTO mode as provided in these rules, except as modified in this chapter.
- (5) Whenever train becomes inoperative or is under control of Train Operator or Roving Attendant due to any reason, all rules of train operation shall apply except as modified in this chapter.
- (6) In DTO or UTO mode, the Traffic Controller shall use all information received at OCC to monitor and ensure smooth operation of trains.

- (7) Other controllers of OCC shall provide necessary support as per special instructions.
- (8) Whenever required, the Traffic Controller shall use remote commands to handle unusual occurrences and call the Roving Attendant or any other authorised person from the nearest location.
- (9) Entry of any personnel including the Roving Attendant to trackside shall be as per special instructions.

88. Signals and Train Controls—

- (1) CBTC or any other technology may be used to provide moving block of system for metro lines.
- (2) Running lines shall be equipped for bi-directional cab signalled operation.
- (3) Moving block system of working shall be used as system of working for driverless lines.
- (4) Driving console shall normally be in covered condition in driverless mode.
- (5) The Roving Attendant when called, shall open driving console and wait till it resumes display as specified in the sub-rule (1) of rule 17.
- (6) Fixed signals, where provided shall be as specified in rules 17 and 18.

89. Working of signals and points — Signal and points shall be controlled from workstations or by signalling system for safe passage of trains, as specified in rule 19.

90. Control of signalling —

- (1) All revenue trains shall be equipped with on board signalling and train control equipments, namely:-
 - (a) CBTC trains operating in CBTC territory shall operate under the protection of the signalling and train control system and the train shall be capable of being controlled manually or self-driven in UTO mode with Roving Attendant on board where required as per special instructions;
 - (b) trains not equipped with CBTC equipment or trains with inoperative train-borne CBTC equipment that are operating in CBTC territory shall operate as per rules 22 and 33 and special instructions;
 - (c) signalling system in emergency situations may permit reversal of a train which has not completed its authorised movement as per special instructions;
 - (d) equipments to perform necessary functionalities as specified in rule 20, shall be provided at stations and on the running lines;
 - (e)
 - (i) UTMS shall normally not command brakes for a train to stop in the neutral section;
 - (ii) the signalling system may not allow CBTC equipped trains to enter a line which is not provided with traction power;
 - (iii) UTMS may provide facility to control entry of train to a section with power failure, where such information is transferred from SCADA to signalling system.
- (2) Trains in underground line—
 - (a) Signalling system may permit more than one train at a time in one tunnel ventilation section, by OCC as per operational requirement, but the Traffic Controller may allow more than one train, as per special instructions for trains in UTO or DTO mode by giving suitable commands at OCC;
 - (b) signalling system shall generate alarm, when train stops more than a predefined time in tunnel ventilation section and the tunnel ventilation fans shall start

automatically, where provided for congestion mode based on such alarm or to be started manually when required;

- (c) the Traffic Controller shall monitor all such trains;
- (d) the location of the train may be transferred by signalling and train control to tunnel ventilation system.
- (e) The auxiliary systems controller may operate tunnel ventilation system as per special instructions.

(3) Safety communication—

All communications between OCC, Roving Attendant, Station Controller, maintenance staff of electrical, rolling stock, signal and telecommunication and track structures department and other pertaining to movement of trains shall be recorded with time stamping, and preserved for incident analysis and training as specified in special instructions.

(4) Running lines—

- (a) Normal operation of the train running on automatic train protection is monitored by UTMS or from the indications on the Train Operator's console, when in ATO or ATP mode and each main running line is duly signalled for operation;
- (b) fixed signals shall be provided at approach to all points and crossing of the interlocked area and certain other locations as per operational requirement.

(5) Stations—

- (a) Station Controller's work-station where provided shall have the capability of setting routes within the area of control and of setting individual points and other functions as assigned;
- (b) train control and signalling in UTO lines shall operate train doors and PSDs wherever provided, start the train from station and stop at stopping point of next station.

(6) Depots—

- (a) Depot tracks may be equipped with full complement of UTMS as per operational requirements and trains shall be automatically restricted to a maximum speed of twenty-five kilometres per hour in non-ATP areas;
- (b) movement within depot shall be in UTO or non-UTO mode depending upon facility provided;
- (c) fixed colour light signals or position light type signals or other means shall be provided for trains with inoperable on board equipment or non-equipped vehicles;
- (d) all movement to and from depot to mainline shall be controlled by signalling;
- (e) when the train is in non-ATP or ATP or ATO mode, verbal instructions and acknowledgement may be exchanged with Train Operator or Roving Attendant for movement between ATP and non-ATP area or between depot and main line on train radio as per special instructions;
- (f) staff protection key and notice and signages in the depot may be suitably placed for entry to UTO area of depot for safety of staff.

91. Train detained on line —

- (1) In case a train operating in UTO mode comes to a stop on running line, the Traffic Controller based on the alarm or message or other information shall attempt to move or restart the train in UTO mode and if the train does not start in UTO mode, the Traffic

Controller shall instruct the Roving Attendant or the ETO to take over the train and make the suitable announcement in the train.

- (2) The Traffic Controller shall check his indications and if he is satisfied that there is no train ahead, he may instruct the Roving Attendant to proceed based on the indications available and in case of non-ATP mode, the Roving Attendant shall exercise utmost vigilance so that he may stop short of any obstruction.
- (3) In case after travelling some distance in ROS mode or RM mode, a proceed indication is received by the train, the train shall automatically upgrade to the coded manual mode of control and the Roving Attendant may select UTO mode with permission from the Traffic Controller.
- (4) In case the cause of the problem is confirmed as a track side equipment malfunction, the Traffic Controller shall arrange to man all the subsequent trains in the affected section.
- (5) The Traffic Controller may instruct subsequent trains without waiting for the Roving Attendant to seek instructions, provided the preceding train has passed through the affected section and reported resumption of coded manual mode.

92. Failures of cab signalling and telecom equipment (s) —

- (1) Failure of any cab signalling or its associated equipments shall result in indication and or alarm to OCC and in most cases the train may still continue to move normally.
- (2) In case ATP system fails, irrevocable emergency brakes shall apply.
- (3) Traffic Controller on getting equipment failure alarm or message shall attempt to release the brakes and move the train in UTO mode as per special instructions and in case, the train does not start after EB release, the Traffic Controller with assistance from concerned controller shall attempt for rectification as per special instructions.
- (4) The Traffic Controller with assistance of concerned controller shall verify as far as practicable from alarms or indications at OCC that the problem is not caused by a track side fault or another train and if no cause is apparent, he shall instruct the Roving Attendant, if on-board, to take over train and try to rectify fault.
- (5) In case the Roving Attendant is not on-board then Traffic Controller shall arrange for a Roving Attendant or ETO to board the train as per special instructions and in case it is not possible to rectify the fault, then the Roving Attendant shall select ATP or non-ATP mode as directed by the Traffic Controller and move ahead exercising utmost vigilance so that he may stop short of any obstruction.
- (6) In case the train still does not move, the Traffic Controller shall authorise the Roving Attendant to operate the Cut-Out switch and to proceed in Cut-Out mode of control to a suitable station, exercising utmost vigilance so that he may stop short of any obstruction.
- (7) Passengers shall be detrained as per special instructions.
- (8) In case of failure of train radio communication to OCC, alarm shall be generated and the train may be allowed to work upto the next station in UTO mode, where the Roving Attendant shall be called by the Traffic Controller to board the train and the train shall then be driven as per special instructions.

93. Failure of fixed signals —

- (1) In case a fixed signal is not displaying any aspect or displaying red aspect due to lamp failure, an indication or alarm shall be displayed at OCC.
- (2) Train in UTO mode may cross the failed signal and if the train stops, the Traffic Controller shall move the stopped train with remote command(s) in case of OFF aspect lamp failure, as per special instructions.
- (3) All the concerned staff shall then follow the provisions of rules 91 and 92 or follow special instructions for UTO mode.

94. Failure of route setting —

- (1) In case a route through an interlocked area cannot be set automatically or by manual control from the OCC, the control of the area shall be passed to the local station control room.
- (2) In case the route cannot be set, the Traffic Controller shall secure the train and instruct the Roving Attendant to take over the train.
- (3) The Roving Attendant, if required shall operate in ROS or RM mode and all concerned staff shall follow provisions of sub-rules (3) to (5) of rule 24.
- (4) In case after some distance the train receives proceed indication while running in ROS Mode, the train shall upgrade to Coded Manual Mode. The Roving Attendant shall inform the OCC and change mode as per instructions.
- (5) On rectification of failure of route setting, at least one train has to cross the affected section in Coded Manual Mode before starting the UTO operation.
- (6) In depot, the depot controller shall instruct the Roving Attendant to take over the train and to proceed in RM or ROS Mode, if —
 - (a) point indications for all points in the route show the points as set and locked in the correct position and blocked as per special instructions;
 - (b) In case points, for which indications are not available, have to be traversed, the depot controller shall ensure that the points are set manually and secured in correct position with the clamp and padlock before authorising the train to proceed.

95. General —

- (1) No person shall perform the duties of a Roving Attendant unless he is in possession of a valid certificate of competency and medical fitness.
- (2) No Roving Attendant shall be booked to work a train until he has become aware of the road and signed a certificate, stating that he is fully acquainted with it and for this purpose, he shall be booked for minimum three round trips including one trip during night, before being put to work the train independently.
- (3) A Roving Attendant who has not operated train on a section for one year or more should be given road learning trips to refresh his knowledge as under:

DURATION OF ABSENCE	NUMBER OF ROAD LEARNING TRIPS
1 year	1 round trip
Over 1 year	3 round trips

- (4) Traffic Controller shall issue remote command for lighting two red lights (flashing) at each end of the stationary train waiting to be rescued.

96. Service regularity;

ATS shall operate and regulate train services as per time table or based on manual commands by Traffic Controller.

- (1) The Traffic Controller shall monitor train operations and take appropriate steps to reduce impact of a disruption.
- (2) Signalling and train control system shall issue departure orders for all scheduled trains.

97. Speed and working of trains—

- (1) Every train shall be run on each line of metro railway within limits of speed as specified in approved special instructions.
- (2) The maximum train speed when passing through a station platform shall be as per the approved special instructions, and in case it is not possible to close any Platform Screen Door or Platform Edge Door, where provided, while receiving, dispatching or passing any

train, the speed of the train shall be restricted to twenty kilometre per hour and an audible warning shall be sounded by the train while entering or leaving the platform.

98. Caution order—

- (1) The caution order shall be issued to Train Operator or Roving Attendant as per special instructions.
- (2) The Traffic Controller and the Station Controller shall ensure that the caution order for speed restriction has been suitably incorporated in the CATC system.

99. Train staffing—

Staff on board shall be as per rule 87.

- (1) The Roving Attendant shall, at all times when on duty, be in possession of equipments as per special instructions.
- (2) The Roving Attendant, when reporting for duty shall examine any notice issued and caution order(s) for his guidance and in particular those which require his special attention on the specific day and line.
- (3) The Roving Attendant shall keep two pairs of spectacles he is required to wear under medical advice.
- (4) The Roving Attendant shall undergo breathalyser test at the time of sign-on and sign-off.
- (5) When the train is required to be started in non UTO or DTO mode from depot itself, the Roving Attendant shall perform his duties as Train Operator.
- (6) In DTO or UTO mode, in the event of Roving Attendant becoming incapacitated while on duty on a train, he shall, if capable, inform the Traffic Controller who shall then inform the Station Controller at the next station, as under,-
 - (a) in case the train is under DTO or UTO mode, the train shall arrive at the next station under automatic control and the on-board Roving Attendant may be relieved at that station for medical attention.
 - (b) The Station Controller, having competency certificate to drive train, may take over as Roving Attendant as per the advice of OCC and the senior most authorised person qualified in the duties of the Station Controller shall be in-charge of the station during his absence.

100. Train defects—

- (a) No train with defective safety equipments, cab signalling, leading cab controls, interior lighting, ventilation, brakes or doors, including emergency doors, shall remain in passenger service, and it shall be removed from service at the earliest as per special instructions.
- (b) Failure of the equipments specified in sub-clause (a), shall generate indications and/or alarms and/or messages at OCC for above defects.
- (c) Remote commands where provided, shall isolate the defective equipments in safe manner.
- (2)
 - (a) In case traction power is lost on any motor car or bogie and the reduction in speed does not cause delay to following trains, the train need not be withdrawn from service and the OCC shall receive alarm of failure indicating the number of bogies isolated for traction.
 - (b) OCC may then decide and either keep train in service or detrain passengers and work train empty as per special instructions;
- (3) Mechanical defects—
 - (i) In the event of an axle lock, TCMS or TIMS, shall generate alarm(s) to OCC so as to ensure safe operation as per special instructions;
 - (ii) In case the Roving Attendant discovers that the train cannot be driven or braked from the leading end cab, the passengers shall be detrained as per the provisions

specified in sub-rules (5) and (6) of rule 30 and train to be operated as per special instructions.

(4) Failure of TCMS or TIMS—

- (i) On failure of a unit of sub-system having redundant configuration for monitoring and control such as TCMS or TIMS, the train shall continue its journey upto the next station and failure shall result in suitable alarm for OCC;
- (ii) the OCC shall take immediate steps to call for a Roving Attendant and in case of second failure of the above sub-system, the Roving Attendant shall operate train as per special instructions and under guidance of OCC.

(5) Brake defects—

- (i) Failure of brakes to apply or to release in terms of bogie count shall be indicated as an alarm at OCC by the TCMS or TIMS;
- (ii) in case of failure of brakes to apply or to release in one bogie, the train may be allowed to work with or without any speed restriction as per special instructions, upto the next station where the OCC shall call for Roving Attendant;
- (iii) whenever failure of brake to apply or to release in terms of bogie count is such that one more bogie failure affects designed brake effort, the train shall be withdrawn at the end of the trip as per special instructions;
- (iv) However, in no case, the train should be moved without release of all the brakes.

(6) Door failures—

In case of train door failure, the train shall be worked as per clause (ii) of sub-rule (10) of rule 30 and remote commands where provided, shall be used as per special instructions.

(7) Air conditioning and lighting—

- (1) On failure of an air conditioning unit, alarm or message shall be sent to OCC by the TCMS or TIMS, so that train receives prompt attention to relieve hardship to passengers;
- (2) on failure of main car lighting on one or more cars, message shall be sent to OCC for prompt attention and the train may continue in passenger service, provided the emergency lighting is working satisfactorily. If all main train lightings and emergency lighting both fail on the same car, alarm shall be sent to OCC and passengers maybe detained at the next station and the train withdrawn from service as per special instructions.

(8) Failure of CCTV—

In case of failure of on-board CCTV communication to OCC, alarm shall be generated and the train may be allowed to work upto the next station in UTO mode, where the Roving Attendant shall be called by Traffic Controller.

101. Examination of trains —

- (1) UTMS shall perform wake-up test of each train before being offered for passenger service.
- (2) Wake-up test examination shall ensure that the following functions of the train are working correctly and in particular safety devices, namely: —
 - (a) cab signalling;
 - (b) safety brake circuits including brake system;
 - (c) train radio communication;
 - (d) CCTV of the train, CCTV communication link;
 - (e) head and tail lights;
 - (f) display panel of TCMS or TIMS,
 - (g) horn;
 - (h) miniature circuit breaker;
 - (i) marker light, flasher light;

- (3) These tests shall result into a suitable log, clearly bringing out that the train is fit for passenger service and shall include the rake ID and results of this test.
- (4) The competent staff shall perform the safety test as specified in sub-rule (2) of rule 31, whenever automatic wake-up test is not performed for whatever reason and follow safety certificate process as specified in sub-rule (3) of rule 31.
- (5) The fitness of train cameras shall be checked as per special instructions.
- (6) The train shall not be certified fit for UTO operation, if the train cameras are not fit as specified in sub-rule (5), but the train may still be operated in non-ATP or ATP or ATO or DTO Mode as per special instructions.
- (7) UTMS shall set route and induct fit trains(s) to main line for passenger service and the first train in UTO line may perform service as per special instructions.
- (8) For trains not fit, the OCC or depot control centre shall take steps for maintenance or induction as per special instructions.
- (9) Any other item including emergency doors, isolation cock ties and all other ties, physical check of brake gears and all seals as required may be checked manually by the rolling stock supervisor as specified under special instructions.

102. Platform doors—

- (1) UTO lines shall be planned with platform doors or intrusion detection system to limit access to track.
- (2) Kinds of platform doors and indications shall be as per the provisions of rules 34 and 37.

103. Normal working of platform doors—

- (1) Platform Screen Doors shall be automatically opened and closed by UTMS.
- (2)
 - (a) In UTO mode when the train is docked properly and permitted by the ATP door release,-
 - (i) the train doors shall open automatically on the correct side;
 - (ii) the Platform Screen Doors, where provided shall open automatically on the correct side;
 - (b) When the train is not docked properly but stops within a defined limit with all doors still on the platform, the train may reverse or move ahead automatically, so as to align the train at normal stopping position.
 - (c) In case a UTO train stops outside the defined limit as specified in clause (b) of sub-rule (2), the Traffic Controller shall take suitable steps in accordance with sub-rule (11) of rule 32.
- (3) In case a train in DTO or UTO mode does not dock automatically at platform, then the Traffic Controller may allow the Roving Attendant to take suitable steps in accordance with sub-rules (8), (9), (10) and (11) of rule 32.

104. Abnormal working and emergency usage of doors—

- (1) Provision of platform supervisor booth may be made and manned as per operational requirement.
- (2) Platform Screen Doors shall give suitable alarms or message or indications to OCC and locally in case of failure of door and speed restrictions or other provisions as specified in rule 36 shall be automated or implemented by OCC and station staff.

105. Responsibilities of Station Controller—

- (1) In UTO mode of operation, Station Controller shall be vigilant for any alarm or message indication of abnormal working.
- (2) He shall also coordinate with crew controller and Traffic Controller for requirement of Roving Attendant.
- (3) Station Controller shall continue to bear responsibilities as specified in rule 39.

106. Emergency evacuation of station—

- (1) The Station Controller and station staff shall perform the following duties in addition to those specified in rule 46, namely:-
 - (i) in case of a fire at station, the Traffic Controller shall be informed of the fire and announcement to that effect may be made by the OCC in the trains and station for clear information to the passengers;
 - (ii) the Station Controller shall also decide depending upon the extent of fire, namely:-
 - (a) to request the Traffic Controller to skip train services at the affected platforms, giving the details thereof;
 - (b) in case the fire is at the concourse level, the train may be stopped to allow passengers at the station to board only, so that passengers are evacuated easily and authorised person in OCC shall make the announcement in the trains to inform passengers not to alight from the train at the affected station;
 - (c) In case the fire is extensive, the trains should not approach the station, and the Station Controller shall inform the Traffic Controller to take steps to prevent the trains coming towards the affected station, to stop or hold at the previous stations for evacuating the passengers there and this should prevail till normalcy is restored to the affected station.
- (2) Automated announcement may be made in case of detection of smoke and/or fire.
- (3) Alarms and messages shall be generated by fire system and sent to station and OCC and the Traffic Controller and Station Controller to take action based on such alarms as per special instructions.

107. Duties of station and OCC staff during unusual occurrences—

OCC and station staff shall perform the following duties in addition to those specified in these rules, namely:-

- (i) a full record of events and actions shall be entered in the log by OCC staff and all system generated alarms, events and messages including video recording shall be preserved;
- (ii) OCC staff shall monitor all alarms, messages and indications for any unusual occurrences such as obstruction, passenger emergency alarm (PEA), fire or smoke alarms from train;
- (iii) OCC shall monitor view from external cameras, if any, on train;
- (iv) whenever the train doors are obstructed or the "doors closed" indication is not received, the Traffic Controller shall issue remote command(s) to close door(s), as necessary and instruct station staff for assistance, if required;
- (v) in case a passenger emergency alarm is operated in the train, the nominated staff in the OCC shall use video communication and try to establish the reason for the operation of alarm and unless there is a clear and immediate danger to the train and passengers, he shall allow the train to continue to the next station before taking any action;
- (vi) the nominated staff in OCC shall communicate with the Station Controller of the station at which the train stops about operation of the passenger emergency alarm;
- (vii) in case of no response for operated emergency alarm from the passenger, the nominated staff in OCC shall view the CCTV cameras of the train and may use this information to reset passenger emergency alarm;
- (viii) during evacuation, the Traffic Controller shall ensure that no train enters the affected area.

108. Train stopped between stations—

- (1) In case a train stops between stations, the signalling shall relay suitable alarm or message to OCC for further action to be taken by the Traffic Controller or rolling stock controller and the OCC shall facilitate necessary announcement in the train.
- (2) In case, the Traffic Controller is not able to isolate a defect on the train and is unable to start the train remotely, he shall instruct the Roving Attendant(s) and follow sub-rule (1) of

rule 50 for rescue of train and shall issue remote commands for lighting two red lights at each end of the stationary train waiting to be rescued.

(3) Traction power lost—

- (a) In case the traction power is lost, the information may be transferred from SCADA to train control and signalling system and the Traffic Controller shall follow sub-rule (2) of rule 50.
- (b) The Traffic Controller shall secure the train before start of evacuation and ensure that no train enters the affected area till the affected area is fit for operation.
- (c) The traction power controller shall also use train camera, where provided, for ascertaining the physical condition of the overhead equipments.

(4) Train unable to move due to derailment or mechanical failure—

- (a) In case a train cannot be moved as a result of derailment or axle lock leading to possible infringement, the train shall switch-on the flasher light and send alarm to the OCC and Traffic Controller to stop all the trains in the affected area;
- (b) the OCC shall take necessary action to restore train movement as per special instructions;
- (c) in case, there is no Roving Attendant on board, the Traffic Controller shall release electrical hold, if provided, on evacuation door, after all the conditions specified in special instructions are fulfilled, and ask passengers,-
 - (i) to wait for arrival of metro staff, or
 - (ii) to self-evacuate, in case of time critical emergencies (e.g. fire);
- (d) the movement of trains on adjacent tracks in DTO or UTO mode and shutdown of traction power on same or adjacent line shall be as per special instructions and the Traffic Controller shall follow provisions of sub-rule (3) of rule 50.
- (e) First train in the section after rescue of the failed train may run in RM mode as per special instructions.

(5) Evacuation of passengers—

To assist evacuation of passengers for the conditions specified in rule 50, the Roving Attendant(s) may be called and the Traffic Controller shall follow sub-rules (4) to (10) of rule 50.

109. Train divided—

- (1) In case, the train is divided, irrevocable emergency brakes shall be applied and TCMS or TIMS shall raise an alarm or give message to OCC.
- (2) In case, a train is stopped by an irrevocable emergency brake application and cab signalling indications are normal, the Traffic Controller shall check the alarms received from train, examine all indications coming from the concerned train to ascertain the cause, if indication of faults in multiple circuits affecting the whole train or rear cars of the train are present, the train shall not be moved, until it has been verified that the train is complete and coupled.
- (3) After the verification of the train integrity under sub-rule (2), the Roving Attendant, if available on-board may make appropriate isolations and proceed after obtaining permission from the OCC.
- (4) In case the train is not divided, but is still unable to move on its own, the Traffic Controller shall issue remote command for lighting two red lights at each end of the stationary train and the train shall be worked by Roving Attendant in accordance with sub-rule (1) of rule 50.
- (5) In case the Roving Attendant is not on-board, then the Traffic Controller shall arrange for the same.

- (6) In case, the train is found to be parted, the Traffic Controller shall first satisfy himself that no passenger on-board has been injured or has fallen from the train and announcements shall be done to guide passengers for self-evacuation as per special instructions. The Traffic Controller shall also arrange for Roving Attendant(s) and follow provisions of rule 51.

110. Unusual occurrences—

- (1) For trains: UTMS shall generate alarms and messages from the train for smoke or fire or obstruction and send these to OCC and on receipt of any fire or smoke alarm, message or information from other sources, the Traffic Controller shall follow the provisions of sub-rules (3) and (4) of rule 52 and follow clause (c) of sub-rule (4) of rule 108, for self evacuation.
- (2) For stations: For any smoke or fire at station, the Traffic Controller shall follow sub-rules (6) to (9) of rule 52.
- (3) For tunnels: In case of smoke or fire in the tunnel and or train, the Traffic Controller and auxiliary system controller shall follow sub-rule (5) of rule 52 for supply of fresh air and extraction of smoke and special instructions, using information from cameras and information from other sources.

111. Flooding—

- (1)
 - (a) Any metro railway employee or authorised person or a Roving Attendant on the train or on-ground or competent staff or Station Controller or the other member of the staff or any alarm received by Station Controller, who observe water accumulating on the track or receives alarm of water accumulation on the track, shall report to the Traffic Controller giving as much detail as possible with respect to location, distance of track affected and approximate level of water with respect to the rail;
 - (b) train front camera where provided, monitoring at the OCC shall also be used to get indication about water accumulation on the track.
- (2) The Traffic Controller shall apply TSR of twenty-five kilometre per hour on the affected area, if the water level is below the level of rail fastenings.
- (3) In case, the water level rises above rail fastenings, passenger train service shall only be permitted under special instructions.

112. Other unsafe conditions—

- (1) In case of report of any unsafe conditions specified in rule 54, the Traffic Controller shall take action to minimise the effect and remove the cause.
- (2) The Roving Attendant, if available on the train shall report the unsafe condition to Traffic Controller as per provisions of sub-rule (2) of rule 54.
- (3) In the event of an earthquake, the Traffic Controller shall stop all the trains immediately by relevant command(s) from OCC and also by instructing Roving Attendant(s), if available on board to stop the train and after the earthquake has subsided, the Traffic Controller shall take action as per sub-rule (3) of rule 54.
- (4) Traffic Controller shall hold trains at platform(s), control train(s) movement to next station including stopping of all trains on the line as required or in affected area in case of unsafe conditions.

113. System of Working – UTMS with moving block, —

- (1) CATC system of working shall be adopted for driverless trains.
- (2) Moving block shall use CBTC or other proven technology for communication between wayside equipment and on board train equipment on the metro railway for movement of trains between stations and between depot and the main line.
- (3) CBTC system is a continuous, automatic train control system utilizing high-resolution train location determination independent of track circuits; continuous, high-capacity, bidirectional train-to-wayside data communications; and train-borne and wayside

processors capable of implementing ATP functions, as well as optional ATO and ATS functions.

- (4) For reversing a train between two stations, procedure as specified in the special instructions shall be followed.
- (5) In addition to systems of working as specified in Chapter 8 of these rules, UTMS of working provides for UTO or DTO modes with capabilities as defined in sub-rule(2) of rule 87.
- (6) CBTC system used shall provide safety and other functions as specified in rules 56 to 62 and also enable UTO or DTO mode of operation.
- (7) Mode of operation in the depot shall be UTO, DTO, ATO, ATP, ROS or RM.

114. Single Line working—

- (1) Running lines and depot lines except sidings, workshop etc., which are not frequently used, may be equipped with bi-directional driverless signalling and train control system and single line working shall be as specified in rules 63, 64 and 65.
- (2) Trains shall work under normal signalling including UTO or DTO in the normal direction of travel and in the reverse direction of travel, if signalling system permits, the train may work under UTO mode or DTO mode, AM or Coded Manual Mode.
- (3) Other modes like RM Mode, ROS Mode or Cut Out Mode may also be used for train operation in the reverse direction of travel as specified in rules 63, 64 and 65, under special instructions.

115. Permanent way and works—

In addition to the provision of rules 66 to 71, the track health monitoring shall be ensured either by suitable systems or special instructions.

116. Power supply and traction management—

In addition to the provisions as specified in rules 72 to 86, information about no power zone and power shut down on running lines shall be transferred to signalling and the train control system so as to minimize the disruption to UTO trains.

CHAPTER – XIII

OTHER SYSTEMS OF WORKING

117. Application of Special Procedure—

Notwithstanding anything contained in these rules, special procedures framed under approved special instructions shall apply to the initial stage of operation or whenever such a condition arises on any section of the metro railway.

Explanation —For the purpose of this rule, the expression “initial stage” means the period in which any section of the metro railway shall be opened without ATP or when signalling and train control is not available from the OCC.

SCHEDULE I

[See rule 81(2)]

(For 25 kV AC Overhead traction)

सावधान

CAUTION

25000 VOLTS

SCHEDULE I

[See rule 81(2)]

(For 750 V DC Traction)